1 Explanation certificate: churn100-s2-c1

This section is the explanation certificate of the churn100-s2-c1 network. It is rendered from the canonical JSON record `churn100-s2-c1.json` (schema 1.4.0, SHA-256 `e92cc46d5657fbeb...`), which also holds every evidence payload, the full graphs with weights, and the explanation record. Table 1 summarises the network and its coverage.

Dataset	churn_onehot
Experiment	churn100-s2, seed 2
Genome	`f188e353-beb3-4b23-9e19-b64aba8cb5ba` (NEAT key 24636, generation 99)
Explanation	`6eb705c3-d493-4a02-ab0a-6e01b8c5d696` (103 operations)
Train AUC / accuracy	0.9009 / 0.9008 (n = 4000)
Test AUC / accuracy	0.8756 / 0.901 (n = 1000)
Output activation	sigmoid
Phenotype	94 nodes (71 inputs, 22 hidden, 1 output), 95 enabled edges
Explained model state	116 nodes, 100 enabled edges
Refactoring operations	add_identity_node 2, consolidate_node 3, split_node 17
Function preservation	max $\lvert\Delta\,\text{output}\rvert = 6.66 \times 10^{-16}$ on the training split
Annotations	10 (8 leaf, 2 composition)
Structural coverage	1 (115 of 115 candidate nodes; 24 Null)
Compositional coverage	1
Uncovered nodes / edges	0 / 0
Missing compositions	0
Full coverage	yes
Drafted by	AI-assisted draft
Verified by	unverified
Record hash (SHA-256)	`e92cc46d5657fbeb...`

Table 1: Explanation certificate summary: churn100-s2-c1.

Drafter's summary. logit P(churn) = -1.05 + a sparse linear backbone over 40 inputs (day minutes +2.04 per SD, voicemail plan -0.90, international plan +0.76, international charge +0.71, voicemail messages +0.62, ...) + a service-call x low-day-usage gate + a 4+-calls hinge + an evening-minutes hinge + exactly linear voicemail and state-offset units, all additive in the logit. The gate is a low-day-minutes hinge whose kink moves up with each customer-service call (median 147 day minutes at 0 calls, 220 at 4, 276 at 6), so with the direct day-minutes edge the logit is V-shaped in day minutes and frequent callers with low usage churn. Service calls act only through hinges. Train AUC 0.901, accuracy 0.901.

What resisted explanation. Nothing structural. Hidden nodes 2627 and 9555 (the always-on service-call encoding) were split so the near-dead state:2/state:44 unit 252 could be its own leaf; 161 and 3861 are identically 0 on train (dead). The gate under-predicts churn for frequent callers with low day usage (CSC>=4 and day minutes <160: 113 rows, model 0.66 vs empirical 0.88) and over-predicts it for low-usage customers with few calls (CSC<4, day minutes <120: 476 rows, 0.08 vs 0.05). closed_form_max_abs_err 0.031 is 3-significant-figure rounding on state one-hots (standardised values up to 8); the numeric function is exact.

1.1 Plain-language summary

AI-assisted draft for review (unverified). Each sentence cites the claims (C01, C02, ...) that back it; every cited claim was computed on the model and the training split and passed its automatic checks. Magnitudes

are changes in predicted risk for a typical patient; confidence words follow the test-split check and n. The text describes what the model does, not the world.

Summary (drafted). For a typical customer (in West Virginia, area code 415, no international or voice-mail plan, 1 call to customer service and median usage) the model predicts a chance of leaving of about 1%; it adds up seven parts: fixed-weight inputs, service calls, a voice-mail discount, heavy evening use, state units, a daytime-calls unit and a small state unit. [C056] This part adds up most inputs with fixed weights, and daytime minutes count most: they raise a typical customer's chance of leaving steadily, from under 1% at 112 to about 13% at 249 minutes. [C057] An international plan takes the chance from about 1% to about 13%, a voice-mail plan lowers it to under 1%, and more voice-mail messages raise it steadily, from about 1% at 0 to about 5% at 32 messages. [C058, C067, C059] A higher international charge, more evening minutes, a higher evening charge and more night-time minutes also raise it steadily, each by one to two points. [C062, C063, C064, C070] Through it, the state moves the chance from about 1% in West Virginia to about 4% in California and in Washington, and to under 1% in Hawaii, Vermont and Colorado (41 to 80 training customers per state). [C060, C061, C066, C068, C071] This part adds up two parts: customer-service calls combined with low daytime use, and a rise from 4 calls upwards. [C111] Through it, daytime minutes lower the chance only while they stay below about 170 minutes (between 147 and 216 minutes for different customers), in an uneven way: for a typical customer, from about 38% at 91 minutes to about 1% at 352 minutes. [C107] With an evening charge above 17.12 US dollars, the chance is lowest at about 154 minutes of daytime use and higher either side: from about 1% at 112 minutes to about 17% at 249 minutes, against about 2% to about 10% at or below that charge. [C103] Customer-service calls add to it from 4 calls upwards, in an uneven way: for a typical such customer, from about 3% at 0 calls to about 11% at 4 calls. [C108] This part reads the voice-mail plan and a few states: through it, a voice-mail plan lowers a typical customer's chance by less than a point, and customers in states other than Alabama, North Carolina and West Virginia get about 2% instead of about 1%. [C125, C126] Through this part, evening minutes add nothing until about 239 minutes, then lower the chance steadily; it acts on 885 of 4000 training customers, 165 of whom left, and for the typical customer it acts on, the chance is about 5% without it and about 4% with it. [C135, C136] This part reads only the state and moves the chance by less than half a point: up in Arizona and Ohio, down in Nevada. [C138, C139, C140] This part reads the state and daytime calls: being in Tennessee raises a typical customer's chance from about 1% to about 2%, and daytime calls barely change it. [C144, C145] This small part moves the chance by less than a tenth of a point, up in Arkansas and Utah. [C150, C153]

The whole model. The model adds up seven parts into one score: fixed-weight inputs, service calls, a voice-mail discount, heavy evening use, state units, a daytime-calls unit and a small state unit. [C056] *Evidence.* It is made up of these seven parts (each explained below); the score is their sum, which was checked on all 4000 training customers. [C056] Changing one input at a time across the whole model, an international plan takes a typical customer's chance of leaving from about 1% to about 13%, and a voice-mail plan lowers it to under 1%. [C001, C012] More customer-service calls raise it in an uneven way, from about 1% at 0 calls to about 3% at 3 calls. [C008] The model ignores the daytime charge, evening calls, international calls and the night-time charge. [C052, C053, C054, C055]

Fixed-weight inputs. This part adds up most inputs with fixed weights: daytime minutes, an international plan and voice-mail messages raise the chance of leaving, steadily for the minutes and messages, and a voice-mail plan lowers it. [C057, C058, C059, C067] *Evidence.* Through it, daytime minutes take a typical customer steadily from under 1% at 112 to about 13% at 249 minutes, and an international plan from about 1% to about 13%; a voice-mail plan lowers the chance to under 1%. [C057, C058, C067] The state moves it from about 1% in West Virginia to about 4% in California and Washington, and to under 1% in Hawaii and Vermont. [C060, C061, C066, C068] Its formula reproduces the part exactly on all 4000 training customers. [C096]

Service calls. This part raises the chance of leaving with calls to customer service, from 4 calls upwards and in an uneven way, and more so for customers with low daytime use; it adds up two parts. [C111, C108, C107] *Evidence.* It is made of two parts that are added: calls with low usage, and many service calls. [C111] Through it, daytime minutes lower a typical customer's chance only while they stay below about 170 minutes (between 147 and 216 minutes for different customers), in an uneven way, from about 38% at 91 minutes to about 1% at 352 minutes. [C107] Customer-service calls add to it from 4 calls upwards, in an uneven way: for a typical such customer, from about 3% at 0 calls to about 11% at 4 calls. [C108] With an evening charge above 17.12 US dollars, the chance is lowest at about 154 minutes of daytime use and higher either side: from about 1% at 112 minutes to about 17% at 249 minutes, against about 2% to about 10% at or below that charge. [C103]

Calls with low usage. This part raises the chance of leaving for customers with low usage, more so when they call customer service: daytime minutes lower it only while they stay below about 170 minutes (between 147 and 216 minutes for different customers), and calls raise it, both in an uneven way. [C119, C113] *Evidence.* It acts on 3121 of 4000 training customers, 402 of whom left; for the typical customer it acts on, it changes the chance by less than a point. [C123] Through it, a typical customer's chance falls in an uneven way from about 38% at 91 minutes of daytime use to about 1% at 352 minutes, with no further fall past about 170 minutes (between 147 and 216 minutes for different customers). [C119] A higher international charge lowers it only while it stays below about 2.95 US dollars (between 2.95 and 4.34 US dollars for different customers), in an uneven way, from about 3% to about 1%. [C121] Customer-service calls raise it in an uneven way, from about 1% at 0 calls to about 3% at 3 calls. [C113] Its formula reproduces the part exactly on all 4000 training customers. [C124]

Many service calls. This part raises the chance of leaving with calls to customer service, adding nothing until about 3 calls, then steadily, except in West Virginia. [C131, C130] *Evidence.* It acts on 306 of 4000 training customers, 154 of whom left; for the typical customer it acts on, the chance is about 16% without it and about 26% with it. [C132] It is switched off in West Virginia: for the typical customer it acts on, being in any other state takes the chance from about 16% to about 26%. [C130] Its formula reproduces the part exactly on all 4000 training customers. [C133]

Voice-mail discount. This part gives customers with a voice-mail plan a small discount on the chance of leaving. [C125] *Evidence.* Through it, a voice-mail plan lowers a typical customer's chance by less than a point, and customers in states other than Alabama, North Carolina and West Virginia get about 2% instead of about 1%. [C125, C126] Its formula reproduces the part exactly on all 4000 training customers. [C129]

Heavy evening use. This part lowers the chance of leaving a little for customers with heavy evening use: evening minutes add nothing until about 239 minutes, then lower it steadily. [C134] *Evidence.* It acts on 885 of 4000 training customers, 165 of whom left; for the typical customer it acts on, the chance is about 5% without it and about 4% with it. [C136] Through it, evening minutes change nothing until about 239 minutes, then lower a typical customer's chance steadily, from about 1% at 138 minutes to under 1% at 266 minutes. [C134] Its formula reproduces the part exactly on all 4000 training customers. [C137]

State units. This part adds small state offsets: up in Arizona and Ohio, down in Nevada. [C138, C139, C140] *Evidence.* Through it, a typical customer's chance moves by less than half a point: up in Arizona and Ohio, down in Nevada, with Missouri and Massachusetts almost unchanged. [C138, C139, C140, C141, C142] Its formula reproduces the part exactly on all 4000 training customers. [C143]

Daytime calls and Tennessee. This part gives customers in Tennessee a small rise in the chance of leaving; daytime calls barely change it. [C144, C145] *Evidence.* Through it, being in Tennessee takes a

typical customer's chance from about 1% to about 2%, and going from 75 to 125 daytime calls barely changes it. [C144, C145] Its formula reproduces the part exactly on all 4000 training customers. [C149]

Small state unit. This small part barely changes the chance of leaving: it adds less than a tenth of a point in Arkansas and Utah. [C150, C153] *Evidence.* Through it, the chance moves by less than a tenth of a point in Arkansas and Utah, a higher international charge lowers it steadily by even less, and customer-service calls barely move it. [C150, C153, C151, C152] Its formula reproduces the part exactly on all 4000 training customers. [C161]

Worth checking against what is known. Worth checking: the state moves the chance by several points (for example California and Washington about 6% and 5%, Hawaii under 1%), based on 41 to 122 training customers per state. [C003, C004, C016] Worth checking: the model treats a voice-mail plan as lowering the chance of leaving but more voice-mail messages as raising it. [C012, C005] Worth checking: the many-service-calls part is switched off for customers in West Virginia, the state of the typical customer. [C130] Worth checking: daytime minutes, the evening charge and the international charge raise the chance through the fixed-weight inputs but lower it through calls with low usage. [C057, C112, C064, C115, C062, C114]

Claims.

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C001	direction: international calling plan: no international plan to an international plan (whole model) *shape: categorical; highest to lowest: an international plan, no international plan*	1% to 13%	log-odds 2.58, OR 13.2; range across patients 2.58 to 2.58	3620, 380	train; test agrees (AUC gain 0.060 [0.035, 0.086], n=1000)	yes	clearly
C002	direction: total daytime minutes: 112 minutes to 249 minutes (whole model) *shape: dip at 160 minutes; 100% of patients same; verified; risk saturates near 0%*	1% to 13%	log-odds 1.55, OR 4.72; range across patients -0.77 to 3.15	4000	train; test agrees (AUC gain 0.092 [0.059, 0.128], n=1000)	no	clearly
C003	direction: state: West Virginia to California (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee ...*	1% to 6%	log-odds 1.84, OR 6.30; range across patients 1.76 to 1.76	122, 41	train; test disagrees (AUC gain -0.016 [-0.030, -0.004], n=1000)	yes	on the training data only
C004	direction: state: West Virginia to Washington (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee ...*	1% to 5%	log-odds 1.66, OR 5.24; range across patients 1.58 to 1.58	122, 72	train; test disagrees (AUC gain -0.016 [-0.030, -0.004], n=1000)	yes	slightly; on the training data only
C005	direction: voice-mail messages: 0 messages to 32 messages (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 5%	log-odds 1.48, OR 4.37; range across patients 1.48 to 1.48	4000	train; test unclear (AUC gain 0.008 [-0.002, 0.019], n=1000)	yes	slightly; on the training data only
C006	direction: state: West Virginia to Montana (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee ...*	1% to 3%	log-odds 1.17, OR 3.22; range across patients 1.09 to 1.09	122, 75	train; test disagrees (AUC gain -0.016 [-0.030, -0.004], n=1000)	yes	slightly; on the training data only
C007	direction: state: West Virginia to Maryland (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee ...*	1% to 3%	log-odds 1.04, OR 2.83; range across patients 0.96 to 0.96	122, 82	train; test disagrees (AUC gain -0.016 [-0.030, -0.004], n=1000)	yes	slightly; on the training data only
C008	direction: customer-service calls: 0 calls to 3 calls (whole model) *shape: accelerating; 84% of patients same; fallback; risk saturates near 0%*	1% to 3%	log-odds 1.19, OR 3.30; range across patients 0.00 to 2.96	4000	train; test agrees (AUC gain 0.097 [0.060, 0.139], n=1000)	yes	slightly
C009	direction: state: West Virginia to Ohio (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee ...*	1% to 2%	log-odds 0.90, OR 2.45; range across patients 0.82 to 0.82	122, 87	train; test disagrees (AUC gain -0.016 [-0.030, -0.004], n=1000)	yes	slightly; on the training data only
C010	direction: state: West Virginia to Tennessee (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee ...*	1% to 2%	log-odds 0.88, OR 2.41; range across patients 0.80 to 0.81	122, 68	train; test disagrees (AUC gain -0.016 [-0.030, -0.004], n=1000)	yes	slightly; on the training data only
C011	direction: total international charge: 1.84 US dollars to 3.70 US dollars (whole model) *shape: accelerating; 87% of patients same; fallback; risk saturates near 0%*	1% to 2%	log-odds 0.67, OR 1.95; range across patients 0.29 to 0.88	4000	train; test unclear (AUC gain 0.005 [-0.001, 0.012], n=1000)	yes	slightly; on the training data only
C012	direction: voice-mail plan: no voice-mail plan to a voice-mail plan (whole model) *shape: categorical; highest to lowest: no voice-mail plan, a voice-mail plan*	1% to <1%	log-odds -2.68, OR 0.07; range across patients -2.68 to -2.68	2939, 1061	train; test agrees (AUC gain 0.033 [0.015, 0.053], n=1000)	yes	slightly

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C013	direction: state: West Virginia to Connecticut (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 2%	log-odds 0.74, OR 2.10; range across patients 0.67 to 0.67	122, 80	train; test disagrees (AUC gain -0.016 [-0.030, -0.004], n=1000)	yes	slightly; on the training data only
C014	direction: total evening minutes: 138 minutes to 266 minutes (whole model) *shape: several pieces; 100% of patients same; fallback; risk saturates near 0%*	1% to 2%	log-odds 0.98, OR 2.66; range across patients 0.98 to 0.98	4000	train; test agrees (AUC gain 0.009 [0.002, 0.016], n=1000)	yes	slightly
C015	direction: state: West Virginia to Oregon (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 2%	log-odds 0.70, OR 2.01; range across patients 0.62 to 0.62	122, 84	train; test disagrees (AUC gain -0.016 [-0.030, -0.004], n=1000)	yes	slightly; on the training data only
C016	direction: state: West Virginia to Hawaii (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to <1%	log-odds -1.75, OR 0.17; range across patients -1.82 to -1.82	122, 73	train; test disagrees (AUC gain -0.016 [-0.030, -0.004], n=1000)	yes	slightly; on the training data only
C017	direction: total evening charge: 11.76 US dollars to 22.57 US dollars (whole model) *shape: dip at 13.46 US dollars; 75% of patients same; fallback; risk saturates near 0%*	1% to 2%	log-odds -0.05, OR 0.95; range across patients -1.93 to 1.14	4000	train; test agrees (AUC gain 0.012 [0.002, 0.024], n=1000)	no	slightly
C018	direction: total night-time minutes: 136 minutes to 264 minutes (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 2%	log-odds 0.78, OR 2.19; range across patients 0.78 to 0.78	4000	train; test agrees (AUC gain 0.006 [0.0005, 0.012], n=1000)	yes	slightly
C019	direction: state: West Virginia to Vermont (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to <1%	log-odds -1.40, OR 0.25; range across patients -1.47 to -1.47	122, 80	train; test disagrees (AUC gain -0.016 [-0.030, -0.004], n=1000)	yes	slightly; on the training data only
C020	direction: state: West Virginia to Arizona (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 2%	log-odds 0.63, OR 1.87; range across patients 0.55 to 0.55	122, 73	train; test disagrees (AUC gain -0.016 [-0.030, -0.004], n=1000)	yes	slightly; on the training data only
C021	direction: state: West Virginia to Colorado (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to <1%	log-odds -0.96, OR 0.38; range across patients -1.04 to -1.04	122, 78	train; test disagrees (AUC gain -0.016 [-0.030, -0.004], n=1000)	yes	slightly; on the training data only
C022	direction: state: West Virginia to Utah (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 2%	log-odds 0.50, OR 1.65; range across patients 0.42 to 0.46	122, 89	train; test disagrees (AUC gain -0.016 [-0.030, -0.004], n=1000)	yes	slightly; on the training data only
C023	direction: state: West Virginia to New Mexico (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 2%	log-odds 0.50, OR 1.64; range across patients 0.42 to 0.42	122, 72	train; test disagrees (AUC gain -0.016 [-0.030, -0.004], n=1000)	yes	slightly; on the training data only
C024	direction: state: West Virginia to state other than Alaska, Alabama, Hawaii, Massachusetts, Arkansas, Maryland, Missouri, Montana, North Carolina, North Dakota, Nebraska, Arizona, New Hampshire, New Mexico, Nevada, Ohio, Oklahoma, Oregon, Pennsylvania, California, South Dakota, Tennessee, Utah, Virginia, Vermont, Washington, Wisconsin, West Virginia, Colorado, Wyoming, Connecticut (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 2%	log-odds 0.50, OR 1.64; range across patients 0.42 to 0.42	122, 1587	train; test disagrees (AUC gain -0.016 [-0.030, -0.004], n=1000)	yes	slightly; on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C025	direction: state: West Virginia to Massachusetts (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 2%	log-odds 0.49, OR 1.64; range across patients 0.42 to 0.42	122, 86	train; test disagrees (AUC gain -0.016 [-0.030, -0.004], n=1000)	yes	slightly; on the training data only
C026	direction: state: West Virginia to Arkansas (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 2%	log-odds 0.47, OR 1.59; range across patients 0.37 to 0.43	122, 78	train; test disagrees (AUC gain -0.016 [-0.030, -0.004], n=1000)	yes	slightly; on the training data only
C027	direction: state: West Virginia to North Carolina (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 2%	log-odds 0.40, OR 1.49; range across patients 0.33 to 0.33	122, 74	train; test disagrees (AUC gain -0.016 [-0.030, -0.004], n=1000)	yes	slightly; on the training data only
C028	direction: state: West Virginia to North Dakota (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 2%	log-odds 0.40, OR 1.49; range across patients 0.32 to 0.32	122, 68	train; test disagrees (AUC gain -0.016 [-0.030, -0.004], n=1000)	yes	slightly; on the training data only
C029	direction: state: West Virginia to Pennsylvania (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 1%	log-odds -0.40, OR 0.67; range across patients -0.47 to -0.47	122, 56	train; test disagrees (AUC gain -0.016 [-0.030, -0.004], n=1000)	yes	slightly; on the training data only
C030	direction: state: West Virginia to Alaska (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 1%	log-odds -0.39, OR 0.68; range across patients -0.46 to -0.46	122, 56	train; test disagrees (AUC gain -0.016 [-0.030, -0.004], n=1000)	yes	slightly; on the training data only
C031	direction: account length: 48 to 152 (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 1%	log-odds 0.36, OR 1.44; range across patients 0.36 to 0.36	4000	train; test disagrees (AUC gain -0.002 [-0.004, 0.0007], n=1000)	yes	slightly; on the training data only
C032	direction: state: West Virginia to Oklahoma (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 1%	log-odds 0.36, OR 1.44; range across patients 0.29 to 0.29	122, 79	train; test disagrees (AUC gain -0.016 [-0.030, -0.004], n=1000)	yes	slightly; on the training data only
C033	direction: total daytime calls: 75 calls to 125 calls (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 1%	log-odds 0.31, OR 1.36; range across patients 0.31 to 0.31	4000	train; test disagrees (AUC gain -0.003 [-0.006, -8e-05], n=1000)	yes	slightly; on the training data only
C034	direction: state: West Virginia to Virginia (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 1%	log-odds -0.28, OR 0.76; range across patients -0.35 to -0.35	122, 91	train; test disagrees (AUC gain -0.016 [-0.030, -0.004], n=1000)	yes	slightly; on the training data only
C035	direction: state: West Virginia to Wyoming (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 1%	log-odds -0.27, OR 0.77; range across patients -0.34 to -0.34	122, 89	train; test disagrees (AUC gain -0.016 [-0.030, -0.004], n=1000)	yes	slightly; on the training data only
C036	direction: state: West Virginia to Wisconsin (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 1%	log-odds -0.13, OR 0.88; range across patients -0.20 to -0.20	122, 86	train; test disagrees (AUC gain -0.016 [-0.030, -0.004], n=1000)	yes	slightly; on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C037	direction: state: West Virginia to Nevada (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 1%	log-odds 0.24, OR 1.27; range across patients 0.16 to 0.16	122, 77	train; test disagrees (AUC gain -0.016 [-0.030, -0.004], n=1000)	yes	slightly; on the training data only
C038	direction: area code: area code 415 to area code 408 (whole model) *shape: categorical; highest to lowest: area code 415, area code 510, area code 408*	1% to 1%	log-odds -0.15, OR 0.86; range across patients -0.15 to -0.15	2016, 1024	train; test disagrees (AUC gain -0.002 [-0.003, -0.0001], n=1000)	yes	slightly; on the training data only
C039	direction: state: West Virginia to Missouri (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 1%	log-odds -0.07, OR 0.93; range across patients -0.14 to -0.14	122, 80	train; test disagrees (AUC gain -0.016 [-0.030, -0.004], n=1000)	yes	slightly; on the training data only
C040	direction: area code: area code 415 to area code 510 (whole model) *shape: categorical; highest to lowest: area code 415, area code 510, area code 408*	1% to 1%	log-odds -0.13, OR 0.88; range across patients -0.13 to -0.13	2016, 960	train; test disagrees (AUC gain -0.002 [-0.003, -0.0001], n=1000)	yes	slightly; on the training data only
C041	direction: state: West Virginia to South Dakota (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 1%	log-odds 0.48, OR 1.62; range across patients -0.37 to 1.32	122, 66	train; test disagrees (AUC gain -0.016 [-0.030, -0.004], n=1000)	no	slightly; on the training data only
C042	direction: state: West Virginia to New Hampshire (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 1%	log-odds 0.13, OR 1.14; range across patients 0.05 to 0.05	122, 74	train; test disagrees (AUC gain -0.016 [-0.030, -0.004], n=1000)	yes	slightly; on the training data only
C043	direction: total international minutes: 7 minutes to 14 minutes (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 1%	log-odds -0.04, OR 0.96; range across patients -0.04 to -0.04	4000	train; test disagrees (AUC gain -0.0002 [-0.0007, 0.0003], n=1000)	yes	slightly; on the training data only
C044	direction: state: West Virginia to Alabama (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 1%	log-odds 0.04, OR 1.04; range across patients -0.04 to -0.04	122, 105	train; test disagrees (AUC gain -0.016 [-0.030, -0.004], n=1000)	no	slightly; on the training data only
C045	direction: state: West Virginia to Nebraska (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 1%	log-odds 0.09, OR 1.10; range across patients 0.02 to 0.02	122, 72	train; test disagrees (AUC gain -0.016 [-0.030, -0.004], n=1000)	yes	slightly; on the training data only
C046	direction: total night-time calls: 74 calls to 125 calls (whole model) *shape: flat; 100% of patients same; verified; risk saturates near 0%*	1% to 1%	log-odds -0.02, OR 0.98; range across patients -0.02 to -0.02	4000	train; test disagrees (AUC gain -0.0001 [-0.0004, 0.0001], n=1000)	yes	slightly; on the training data only
C047	magnitude: international calling plan: no international plan to an international plan, typical patient *shape: categorical; highest to lowest: an international plan, no international plan*	1% to 13%	log-odds 2.58, OR 13.2	3620, 380	train; test agrees (AUC gain 0.060 [0.035, 0.086], n=1000)	yes	clearly
C048	magnitude: state: West Virginia to California, typical patient *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 6%	log-odds 1.76, OR 5.84	122, 41	train; test disagrees (AUC gain -0.016 [-0.030, -0.004], n=1000)	yes	on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C049	magnitude: voice-mail messages: 0 messages to 32 messages, typical patient *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 5%	log-odds 1.48, OR 4.37	4000	train; test unclear (AUC gain 0.008 [-0.002, 0.019], n=1000)	yes	slightly; on the training data only
C050	magnitude: customer-service calls: 0 calls to 3 calls, typical patient *shape: accelerating; 84% of patients same; fallback; risk saturates near 0%*	1% to 3%	log-odds 0.93, OR 2.52	4000	train; test agrees (AUC gain 0.097 [0.060, 0.139], n=1000)	yes	slightly
C051	magnitude: total international charge: 1.84 US dollars to 3.70 US dollars, typical patient *shape: accelerating; 87% of patients same; fallback; risk saturates near 0%*	1% to 2%	log-odds 0.88, OR 2.41	4000	train; test unclear (AUC gain 0.005 [-0.001, 0.012], n=1000)	yes	slightly; on the training data only
C052	null: model ignores total daytime charge (exact rule)		max change 0.000 (max |change in log-odds|)	4000	train	yes	
C053	null: model ignores total evening calls (exact rule)		max change 0.000 (max |change in log-odds|)	4000	train	yes	
C054	null: model ignores total international calls (exact rule)		max change 0.000 (max |change in log-odds|)	4000	train	yes	
C055	null: model ignores total night-time charge (exact rule)		max change 0.000 (max |change in log-odds|)	4000	train	yes	
C056	structure: made of linear_backbone, service_call_nonlinear, intl_state_unit, eve_minutes_hinge, voicemail_chain, state_offsets, day_calls_state42; added		log-odds n/a, OR n/a	4000	train	yes	
C057	direction: total daytime minutes: 112 minutes to 249 minutes (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	<1% to 13%	log-odds 5.19, OR 179; range across patients 5.19 to 5.19	4000	train; test agrees (AUC gain 0.188 [0.143, 0.234], n=1000)	yes	clearly
C058	direction: international calling plan: no international plan to an international plan (through this part) *shape: categorical; highest to lowest: an international plan, no international plan*	1% to 13%	log-odds 2.58, OR 13.2; range across patients 2.58 to 2.58	3620, 380	train; test agrees (AUC gain 0.060 [0.035, 0.086], n=1000)	yes	clearly
C059	direction: voice-mail messages: 0 messages to 32 messages (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 5%	log-odds 1.48, OR 4.37; range across patients 1.48 to 1.48	4000	train; test unclear (AUC gain 0.008 [-0.002, 0.019], n=1000)	yes	slightly; on the training data only
C060	direction: state: West Virginia to California (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to 4%	log-odds 1.41, OR 4.10; range across patients 1.41 to 1.41	122, 41	train; test disagrees (AUC gain -0.012 [-0.025, -0.0006], n=1000)	yes	slightly; on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C061	direction: state: West Virginia to Washington (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to 4%	log-odds 1.23, OR 3.41; range across patients 1.23 to 1.23	122, 72	train; test disagrees (AUC gain -0.012 [-0.025, -0.0006], n=1000)	yes	slightly; on the training data only
C062	direction: total international charge: 1.84 US dollars to 3.70 US dollars (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	<1% to 3%	log-odds 1.76, OR 5.79; range across patients 1.76 to 1.76	4000	train; test agrees (AUC gain 0.018 [0.006, 0.030], n=1000)	yes	slightly
C063	direction: total evening minutes: 138 minutes to 266 minutes (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 2%	log-odds 1.30, OR 3.66; range across patients 1.30 to 1.30	4000	train; test agrees (AUC gain 0.020 [0.009, 0.032], n=1000)	yes	slightly
C064	direction: total evening charge: 11.76 US dollars to 22.57 US dollars (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 2%	log-odds 1.14, OR 3.12; range across patients 1.14 to 1.14	4000	train; test agrees (AUC gain 0.017 [0.007, 0.027], n=1000)	yes	slightly
C065	direction: state: West Virginia to Montana (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to 2%	log-odds 0.74, OR 2.10; range across patients 0.74 to 0.74	122, 75	train; test disagrees (AUC gain -0.012 [-0.025, -0.0006], n=1000)	yes	slightly; on the training data only
C066	direction: state: West Virginia to Hawaii (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to <1%	log-odds -2.18, OR 0.11; range across patients -2.18 to -2.18	122, 73	train; test disagrees (AUC gain -0.012 [-0.025, -0.0006], n=1000)	yes	slightly; on the training data only
C067	direction: voice-mail plan: no voice-mail plan to a voice-mail plan (through this part) *shape: categorical; highest to lowest: no voice-mail plan, a voice-mail plan*	1% to <1%	log-odds -2.04, OR 0.13; range across patients -2.04 to -2.04	2939, 1061	train; test agrees (AUC gain 0.018 [0.004, 0.034], n=1000)	yes	slightly
C068	direction: state: West Virginia to Vermont (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to <1%	log-odds -1.83, OR 0.16; range across patients -1.83 to -1.83	122, 80	train; test disagrees (AUC gain -0.012 [-0.025, -0.0006], n=1000)	yes	slightly; on the training data only
C069	direction: state: West Virginia to Maryland (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to 2%	log-odds 0.61, OR 1.84; range across patients 0.61 to 0.61	122, 82	train; test disagrees (AUC gain -0.012 [-0.025, -0.0006], n=1000)	yes	slightly; on the training data only
C070	direction: total night-time minutes: 136 minutes to 264 minutes (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 2%	log-odds 0.78, OR 2.19; range across patients 0.78 to 0.78	4000	train; test agrees (AUC gain 0.006 [0.0005, 0.012], n=1000)	yes	slightly
C071	direction: state: West Virginia to Colorado (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to <1%	log-odds -1.39, OR 0.25; range across patients -1.39 to -1.39	122, 78	train; test disagrees (AUC gain -0.012 [-0.025, -0.0006], n=1000)	yes	slightly; on the training data only
C072	direction: state: West Virginia to Pennsylvania (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to <1%	log-odds -0.83, OR 0.44; range across patients -0.83 to -0.83	122, 56	train; test disagrees (AUC gain -0.012 [-0.025, -0.0006], n=1000)	yes	slightly; on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C073	direction: state: West Virginia to Alaska (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to <1%	log-odds -0.82, OR 0.44; range across patients -0.82 to -0.82	122, 56	train; test disagrees (AUC gain -0.012 [-0.025, -0.0006], n=1000)	yes	slightly; on the training data only
C074	direction: state: West Virginia to South Dakota (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to 1%	log-odds -0.73, OR 0.48; range across patients -0.73 to -0.73	122, 66	train; test disagrees (AUC gain -0.012 [-0.025, -0.0006], n=1000)	yes	slightly; on the training data only
C075	direction: state: West Virginia to Virginia (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to 1%	log-odds -0.70, OR 0.49; range across patients -0.70 to -0.70	122, 91	train; test disagrees (AUC gain -0.012 [-0.025, -0.0006], n=1000)	yes	slightly; on the training data only
C076	direction: state: West Virginia to Wyoming (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to 1%	log-odds -0.69, OR 0.50; range across patients -0.69 to -0.69	122, 89	train; test disagrees (AUC gain -0.012 [-0.025, -0.0006], n=1000)	yes	slightly; on the training data only
C077	direction: state: West Virginia to Wisconsin (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to 1%	log-odds -0.55, OR 0.57; range across patients -0.55 to -0.55	122, 86	train; test disagrees (AUC gain -0.012 [-0.025, -0.0006], n=1000)	yes	slightly; on the training data only
C078	direction: state: West Virginia to Missouri (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to 1%	log-odds -0.46, OR 0.63; range across patients -0.46 to -0.46	122, 80	train; test disagrees (AUC gain -0.012 [-0.025, -0.0006], n=1000)	yes	slightly; on the training data only
C079	direction: state: West Virginia to Connecticut (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to 2%	log-odds 0.31, OR 1.37; range across patients 0.31 to 0.31	122, 80	train; test disagrees (AUC gain -0.012 [-0.025, -0.0006], n=1000)	yes	slightly; on the training data only
C080	direction: account length: 48 to 152 (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 1%	log-odds 0.36, OR 1.44; range across patients 0.36 to 0.36	4000	train; test disagrees (AUC gain -0.002 [-0.004, 0.0007], n=1000)	yes	slightly; on the training data only
C081	direction: total daytime calls: 75 calls to 125 calls (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 1%	log-odds 0.32, OR 1.38; range across patients 0.32 to 0.32	4000	train; test disagrees (AUC gain -0.003 [-0.006, -8e-06], n=1000)	yes	slightly; on the training data only
C082	direction: state: West Virginia to Oregon (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to 1%	log-odds 0.27, OR 1.31; range across patients 0.27 to 0.27	122, 84	train; test disagrees (AUC gain -0.012 [-0.025, -0.0006], n=1000)	yes	slightly; on the training data only
C083	direction: state: West Virginia to North Carolina (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to 1%	log-odds 0.26, OR 1.30; range across patients 0.26 to 0.26	122, 74	train; test disagrees (AUC gain -0.012 [-0.025, -0.0006], n=1000)	yes	slightly; on the training data only
C084	direction: state: West Virginia to Ohio (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to 1%	log-odds 0.26, OR 1.29; range across patients 0.26 to 0.26	122, 87	train; test disagrees (AUC gain -0.012 [-0.025, -0.0006], n=1000)	yes	slightly; on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C085	direction: state: West Virginia to Nebraska (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to 1%	log-odds -0.33, OR 0.72; range across patients -0.33 to -0.33	122, 72	train; test disagrees (AUC gain -0.012 [-0.025, -0.0006], n=1000)	yes	slightly; on the training data only
C086	direction: state: West Virginia to New Hampshire (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to 1%	log-odds -0.30, OR 0.74; range across patients -0.30 to -0.30	122, 74	train; test disagrees (AUC gain -0.012 [-0.025, -0.0006], n=1000)	yes	slightly; on the training data only
C087	direction: area code: area code 415 to area code 408 (through this part) *shape: categorical; highest to lowest: area code 415, area code 510, area code 408*	1% to 1%	log-odds -0.15, OR 0.86; range across patients -0.15 to -0.15	2016, 1024	train; test disagrees (AUC gain -0.002 [-0.003, -0.0001], n=1000)	yes	slightly; on the training data only
C088	direction: area code: area code 415 to area code 510 (through this part) *shape: categorical; highest to lowest: area code 415, area code 510, area code 408*	1% to 1%	log-odds -0.13, OR 0.88; range across patients -0.13 to -0.13	2016, 960	train; test disagrees (AUC gain -0.002 [-0.003, -0.0001], n=1000)	yes	slightly; on the training data only
C089	direction: state: West Virginia to Arizona (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to 1%	log-odds -0.10, OR 0.91; range across patients -0.10 to -0.10	122, 73	train; test disagrees (AUC gain -0.012 [-0.025, -0.0006], n=1000)	yes	slightly; on the training data only
C090	direction: state: West Virginia to state other than Alaska, Hawaii, Arkansas, Maryland, Missouri, Montana, North Carolina, North Dakota, Nebraska, Arizona, New Hampshire, Ohio, Oklahoma, Oregon, Pennsylvania, California, South Dakota, Virginia, Vermont, Washington, Wisconsin, West Virginia, Colorado, Wyoming, Connecticut (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to 1%	log-odds 0.07, OR 1.07; range across patients 0.07 to 0.07	122, 2084	train; test disagrees (AUC gain -0.012 [-0.025, -0.0006], n=1000)	yes	slightly; on the training data only
C091	direction: state: West Virginia to Oklahoma (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to 1%	log-odds -0.07, OR 0.94; range across patients -0.07 to -0.07	122, 79	train; test disagrees (AUC gain -0.012 [-0.025, -0.0006], n=1000)	yes	slightly; on the training data only
C092	direction: total international minutes: 7 minutes to 14 minutes (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 1%	log-odds -0.04, OR 0.96; range across patients -0.04 to -0.04	4000	train; test disagrees (AUC gain -0.0002 [-0.0007, 0.0003], n=1000)	yes	slightly; on the training data only
C093	direction: state: West Virginia to North Dakota (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to 1%	log-odds -0.03, OR 0.97; range across patients -0.03 to -0.03	122, 68	train; test disagrees (AUC gain -0.012 [-0.025, -0.0006], n=1000)	yes	slightly; on the training data only
C094	direction: total night-time calls: 74 calls to 125 calls (through this part) *shape: flat; 100% of patients same; verified; risk saturates near 0%*	1% to 1%	log-odds -0.02, OR 0.98; range across patients -0.02 to -0.02	4000	train; test disagrees (AUC gain -0.0001 [-0.0004, 0.0001], n=1000)	yes	slightly; on the training data only
C095	direction: state: West Virginia to Arkansas (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to 1%	log-odds -0.02, OR 0.98; range across patients -0.02 to -0.02	122, 78	train; test disagrees (AUC gain -0.012 [-0.025, -0.0006], n=1000)	yes	slightly; on the training data only
C096	structure: its formula reproduces it on 4000 training patients (largest difference 3.55271e-15)		log-odds 4e-15, OR n/a	4000	train	yes	

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C097	direction: total daytime minutes: 112 minutes to 249 minutes (through this part) *shape: hinge at 170 minutes; 71% of patients same; fallback; risk saturates near 0%; one point with C107 (median fold 170 minutes, typical patient 160 minutes, 10th-90th percentile 147 minutes to 216 minutes, wide)*	15% to 1%	log-odds -3.63, OR 0.03; range across patients -5.95 to -2.04	4000	train; test agrees (AUC gain 0.062 [0.035, 0.095], n=1000)	yes	clearly
C098	direction: customer-service calls: 0 calls to 3 calls (through this part) *shape: accelerating; 84% of patients same; fallback; risk saturates near 0%*	1% to 3%	log-odds 1.19, OR 3.30; range across patients 0.00 to 2.96	4000	train; test agrees (AUC gain 0.097 [0.060, 0.139], n=1000)	yes	slightly
C099	direction: total international charge: 1.84 US dollars to 3.70 US dollars (through this part) *shape: hinge at 2.95 US dollars; 84% of patients same; fallback; risk saturates near 0%; one point with C109 (median fold 2.95 US dollars, typical patient 2.92 US dollars, 10th-90th percentile 2.95 US dollars to 4.34 US dollars, wide)*	2% to 1%	log-odds -1.08, OR 0.34; range across patients -1.46 to -0.88	4000	train; test unclear (AUC gain 0.006 [-0.001, 0.014], n=1000)	yes	slightly; on the training data only
C100	direction: total evening charge: 11.76 US dollars to 22.57 US dollars (through this part) *shape: hinge at 18.11 US dollars; 65% of patients same; fallback; risk saturates near 0%; one point with C110 (median fold 18.11 US dollars, typical patient 13.46 US dollars, 10th-90th percentile 7.97 US dollars to 21.09 US dollars, wide)*	2% to 1%	log-odds -1.19, OR 0.30; range across patients -3.06 to 0.00	4000	train; test unclear (AUC gain 0.009 [-0.003, 0.020], n=1000)	yes	slightly; on the training data only
C101	direction: state: West Virginia to South Dakota (through this part) *shape: categorical; highest to lowest: South Dakota, West Virginia, state other than South Dakota, West Virginia*	1% to 2%	log-odds 0.86, OR 2.36; range across patients 0.00 to 1.70	122, 66	train; test disagrees (AUC gain -0.002 [-0.004, -0.0001], n=1000)	yes	slightly; on the training data only
C102	direction: state: West Virginia to state other than South Dakota, West Virginia (through this part) *shape: categorical; highest to lowest: South Dakota, West Virginia, state other than South Dakota, West Virginia*	1% to 1%	log-odds 0.08, OR 1.08; range across patients 0.00 to 0.00	122, 3812	train; test disagrees (AUC gain -0.002 [-0.004, -0.0001], n=1000)	yes	slightly; on the training data only
C103	interaction: effect of total daytime minutes (112 minutes to 249 minutes) by total evening charge: total evening charge above 17.12 US dollars minus total evening charge at or below 17.12 US dollars; stronger in total evening charge above 17.12 US dollars *shape: dip at 154 minutes; 100% of patients same; verified; risk saturates near 0%*	total evening charge above 17.12 US dollars: 1% to 17%; total evening charge at or below 17.12 US dollars: 2% to 10%	log-odds difference 1.53, OR 4.60	1998, 2002	train; test agrees (AUC gain 0.009 [-0.003, 0.020], n=1000); group difference on test 1.53 [1.35, 1.69]	yes	clearly
C104	interaction: effect of total daytime minutes (112 minutes to 249 minutes) by state: South Dakota minus West Virginia; stronger in West Virginia *shape: dip at 160 minutes; 98% of patients same; verified; risk saturates near 0%*	South Dakota: 5% to 9%; West Virginia: 1% to 13%	log-odds difference -1.09, OR 0.34	66, 122	train; test agrees (AUC gain -0.002 [-0.004, -0.0001], n=1000); group difference on test -1.05 [-1.92, -0.17]	yes	clearly
C105	interaction: effect of customer-service calls (0 calls to 3 calls) by total evening charge: total evening charge above 17.12 US dollars minus total evening charge at or below 17.12 US dollars; stronger in total evening charge at or below 17.12 US dollars *shape: accelerating; 89% of patients same; fallback; risk saturates near 0%*	total evening charge above 17.12 US dollars: 1% to 2%; total evening charge at or below 17.12 US dollars: 1% to 5%	log-odds difference -0.81, OR 0.45	1998, 2002	train; test agrees (AUC gain 0.009 [-0.003, 0.020], n=1000); group difference on test -0.91 [-1.04, -0.76]	yes	slightly
C106	interaction: effect of total	South Dakota:	log-odds difference	66, 122	train; test unclear	yes	slightly; on

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C109	threshold: total international charge: fold at about 2.95 US dollars (stops) *shape: hinge at 2.95 US dollars; 88% of patients same; fallback; risk saturates near 0%; one point with C109 (median fold 2.95 US dollars, typical patient 2.92 US dollars, 10th-90th percentile 2.95 US dollars to 4.34 US dollars, wide)*	3% to 1% (typical patient with a fold)	fold 2.95 US dollars (IQR 2.95 to 3.62); past it log-odds -1.42, OR 0.24; range across patients -2.21 to -1.11	3851, 2752	train; test unclear (AUC gain 0.006 [-0.001, 0.014], n=1000)	yes	slightly; on the training data only
C110	threshold: total evening charge: fold at about 18.11 US dollars (stops) for customers in Colorado *shape: hinge at 18.11 US dollars; 75% of patients same; fallback; risk saturates near 0%; one point with C110 (median fold 18.11 US dollars, typical patient 17.01 US dollars, 10th-90th percentile 7.97 US dollars to 21.09 US dollars, wide)*	2% to <1% (typical patient with a fold (customers in Colorado))	fold 18.1 US dollars (IQR 14.6 to 18.7); past it log-odds -2.12, OR 0.12; range across patients -3.60 to 0.00	71, 34	train; test unclear (AUC gain 0.009 [-0.003, 0.020], n=1000)	yes	slightly; on the training data only
C111	structure: made of csc_ usage_gate, csc_hinge; added		log-odds n/a, OR n/a	4000	train	yes	
C112	direction: total daytime minutes: 112 minutes to 249 minutes (through this part) *shape: hinge at 170 minutes; 71% of patients same; fallback; risk saturates near 0%; one point with C119 (median fold 170 minutes, typical patient 160 minutes, 10th-90th percentile 147 minutes to 216 minutes, wide)*	15% to 1%	log-odds -3.63, OR 0.03; range across patients -5.95 to -2.04	4000	train; test agrees (AUC gain 0.062 [0.035, 0.095], n=1000)	yes	clearly
C113	direction: customer-service calls: 0 calls to 3 calls (through this part) *shape: accelerating; 80% of patients same; fallback; risk saturates near 0%*	1% to 3%	log-odds 1.19, OR 3.30; range across patients 0.00 to 2.96	4000	train; test agrees (AUC gain 0.061 [0.035, 0.091], n=1000)	yes	slightly
C114	direction: total international charge: 1.84 US dollars to 3.70 US dollars (through this part) *shape: hinge at 2.95 US dollars; 84% of patients same; fallback; risk saturates near 0%; one point with C121 (median fold 2.95 US dollars, typical patient 2.92 US dollars, 10th-90th percentile 2.95 US dollars to 4.34 US dollars, wide)*	2% to 1%	log-odds -1.08, OR 0.34; range across patients -1.46 to -0.88	4000	train; test unclear (AUC gain 0.006 [-0.001, 0.014], n=1000)	yes	slightly; on the training data only
C115	direction: total evening charge: 11.76 US dollars to 22.57 US dollars (through this part) *shape: hinge at 18.11 US dollars; 65% of patients same; fallback; risk saturates near 0%; one point with C122 (median fold 18.11 US dollars, typical patient 13.46 US dollars, 10th-90th percentile 7.97 US dollars to 21.09 US dollars, wide)*	2% to 1%	log-odds -1.19, OR 0.30; range across patients -3.06 to 0.00	4000	train; test unclear (AUC gain 0.009 [-0.003, 0.020], n=1000)	yes	slightly; on the training data only
C116	direction: state: state other than South Dakota to South Dakota (through this part) *shape: categorical; highest to lowest: South Dakota, state other than South Dakota*	1% to 2%	log-odds 0.78, OR 2.19; range across patients 0.00 to 1.70	3934, 66	train; test disagrees (AUC gain -0.0007 [-0.002, 0.0001], n=1000)	yes	slightly; on the training data only
C117	interaction: effect of total daytime minutes (112 minutes to 249 minutes) by state: South Dakota minus state other than South Dakota; stronger in state other than South Dakota *shape: dip at 160 minutes; 100% of patients same; verified; risk saturates near 0%*	South Dakota: 5% to 9%; state other than South Dakota: 1% to 13%	log-odds difference -1.26, OR 0.28	66, 3934	train; test agrees (AUC gain -0.0007 [-0.002, 0.0001], n=1000); group difference on test -1.38 [-2.09, -0.66]	yes	clearly
C118	interaction: effect of total evening charge (11.76 US dollars to 22.57 US dollars) by state: South Dakota minus state other than South Dakota; stronger in South Dakota *shape: dip at 18.52 US*	South Dakota: 4% to 1%; state other than South Dakota: 1% to 2%	log-odds difference -0.90, OR 0.41	66, 3934	train; test unclear (AUC gain -0.0007 [-0.002, 0.0001], n=1000); group difference on test -0.60 [-1.23, 0.05]	yes	slightly; on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C121	threshold: total international charge: fold at about 2.95 US dollars (stops) *shape: hinge at 2.95 US dollars; 88% of patients same; fallback; risk saturates near 0%; one point with C121 (median fold 2.95 US dollars, typical patient 2.92 US dollars, 10th-90th percentile 2.95 US dollars to 4.34 US dollars, wide)*	3% to 1% (typical patient with a fold)	fold 2.95 US dollars (IQR 2.95 to 3.62); past it log-odds -1.42, OR 0.24; range across patients -2.21 to -1.11	3851, 2752	train; test unclear (AUC gain 0.006 [-0.001, 0.014], n=1000)	yes	slightly; on the training data only
C122	threshold: total evening charge: fold at about 18.11 US dollars (stops) for customers in Colorado *shape: hinge at 18.11 US dollars; 75% of patients same; fallback; risk saturates near 0%; one point with C122 (median fold 18.11 US dollars, typical patient 17.01 US dollars, 10th-90th percentile 7.97 US dollars to 21.09 US dollars, wide)*	2% to <1% (typical patient with a fold (customers in Colorado))	fold 18.1 US dollars (IQR 14.6 to 18.7); past it log-odds -2.12, OR 0.12; range across patients -3.60 to 0.00	71, 34	train; test unclear (AUC gain 0.009 [-0.003, 0.020], n=1000)	yes	slightly; on the training data only
C123	subgroup: who this part acts on: 3121 of 4000 patients	<1% to 1% (typical patient this part acts on (risk without the part, then with it))	share 0.78; log-odds 2.00, OR 7.39; range across patients 0.18 to 5.02	3121, 879	train; test agrees (AUC gain 0.141 [0.093, 0.190], n=1000)	yes	slightly
C124	structure: its formula reproduces it on 4000 training patients (largest difference 1.77636e-15)		log-odds 2e-15, OR n/a	4000	train	yes	
C125	direction: voice-mail plan: no voice-mail plan to a voice-mail plan (through this part) *shape: categorical; highest to lowest: no voice-mail plan, a voice-mail plan*	1% to 1%	log-odds -0.64, OR 0.53; range across patients -0.64 to -0.64	2939, 1061	train; test unclear (AUC gain 0.002 [-0.004, 0.006], n=1000)	yes	slightly; on the training data only
C126	direction: state: West Virginia to state other than Alabama, North Carolina, West Virginia (through this part) *shape: categorical; highest to lowest: state other than Alabama, North Carolina, West Virginia, North Carolina, West Virginia, Alabama*	1% to 2%	log-odds 0.35, OR 1.42; range across patients 0.35 to 0.35	122, 3699	train; test disagrees (AUC gain -0.0005 [-0.002, 0.001], n=1000)	yes	slightly; on the training data only
C127	direction: state: West Virginia to Alabama (through this part) *shape: categorical; highest to lowest: state other than Alabama, North Carolina, West Virginia, North Carolina, West Virginia, Alabama*	1% to 1%	log-odds -0.10, OR 0.90; range across patients -0.10 to -0.10	122, 105	train; test disagrees (AUC gain -0.0005 [-0.002, 0.001], n=1000)	yes	slightly; on the training data only
C128	direction: state: West Virginia to North Carolina (through this part) *shape: categorical; highest to lowest: state other than Alabama, North Carolina, West Virginia, North Carolina, West Virginia, Alabama*	1% to 1%	log-odds 0.06, OR 1.06; range across patients 0.06 to 0.06	122, 74	train; test disagrees (AUC gain -0.0005 [-0.002, 0.001], n=1000)	yes	slightly; on the training data only
C129	structure: its formula reproduces it on 4000 training patients (largest difference 4.44089e-16)		log-odds 4e-16, OR n/a	4000	train	yes	
C130	direction: state: West Virginia to state other than West Virginia (through this part) *shape: categorical; highest to lowest: state other than West Virginia, West Virginia*	16% to 26% (typical patient this part acts on)	log-odds 0.08, OR 1.08; range across patients 0.00 to 0.00	122, 3878	train; test disagrees (AUC gain -0.001 [-0.003, 0.0002], n=1000)	yes	on the training data only
C131	threshold: customer-service calls: fold at about 4 calls (starts) *shape: hinge at 3 calls; 100% of patients same; verified; risk saturates near 0%; one point with C131 (median fold 4 calls, typical patient 3 calls, 10th-90th percentile 3 calls to 3 calls)*	1% to 1% (typical patient with a fold)	fold 3.02 calls (IQR 3.02 to 3.02); past it log-odds 0.58, OR 1.79; range across patients 0.60 to 0.60	4000, 306	train; test unclear (AUC gain 0.002 [-0.001, 0.006], n=1000)	yes	slightly; on the training data only
C132	subgroup: who this part acts on: 306 of 4000 patients	16% to 26% (typical patient this part acts on (risk without the part, then with it))	share 0.08; log-odds 0.95, OR 2.58; range across patients 0.60 to 1.83	306, 3694	train; test unclear (AUC gain 0.001 [-0.002, 0.005], n=1000)	yes	on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C133	structure: its formula reproduces it on 4000 training patients (largest difference 2.22045e-16)		log-odds 2e-16, OR n/a	4000	train	yes	
C134	direction: total evening minutes: 138 minutes to 266 minutes (through this part) *shape: hinge at 239 minutes; 100% of patients same; verified; risk saturates near 0%; one point with C135 (median fold 239 minutes, typical patient 236 minutes, 10th-90th percentile 239 minutes to 239 minutes)*	1% to 1%	log-odds -0.32, OR 0.73; range across patients -0.32 to -0.32	4000	train; test disagrees (AUC gain -0.004 [-0.009, 0.0002], n=1000)	yes	slightly; on the training data only
C135	threshold: total evening minutes: fold at about 239 minutes (starts) *shape: hinge at 239 minutes; 100% of patients same; verified; risk saturates near 0%; one point with C135 (median fold 239 minutes, typical patient 236 minutes, 10th-90th percentile 239 minutes to 239 minutes)*	1% to 1% (typical patient with a fold)	fold 239 minutes (IQR 239 to 239); past it log-odds -0.54, OR 0.58; range across patients -0.54 to -0.54	4000, 885	train; test disagrees (AUC gain -0.004 [-0.009, 0.0002], n=1000)	yes	slightly; on the training data only
C136	subgroup: who this part acts on: 885 of 4000 patients	5% to 4% (typical patient this part acts on (risk without the part, then with it))	share 0.22; log-odds -0.35, OR 0.70; range across patients -0.77 to -0.05	885, 3115	train; test disagrees (AUC gain -0.004 [-0.009, 0.0002], n=1000)	yes	slightly; on the training data only
C137	structure: its formula reproduces it on 4000 training patients (largest difference 0.0)		log-odds 0.00, OR n/a	4000	train	yes	
C138	direction: state: state other than Massachusetts, Missouri, Arizona, Nevada, Ohio, Wyoming to Arizona (through this part) *shape: categorical; highest to lowest: Arizona, Ohio, Wyoming, state other than Massachusetts, Missouri, Arizona, Nevada, Ohio, Wyoming, Massachusetts, Missouri …*	1% to 1%	log-odds 0.30, OR 1.35; range across patients 0.30 to 0.30	3508, 73	train; test disagrees (AUC gain -0.0003 [-0.001, 0.001], n=1000)	yes	slightly; on the training data only
C139	direction: state: state other than Massachusetts, Missouri, Arizona, Nevada, Ohio, Wyoming to Ohio (through this part) *shape: categorical; highest to lowest: Arizona, Ohio, Wyoming, state other than Massachusetts, Missouri, Arizona, Nevada, Ohio, Wyoming, Massachusetts, Missouri …*	1% to 1%	log-odds 0.21, OR 1.24; range across patients 0.21 to 0.21	3508, 87	train; test disagrees (AUC gain -0.0003 [-0.001, 0.001], n=1000)	yes	slightly; on the training data only
C140	direction: state: state other than Massachusetts, Missouri, Arizona, Nevada, Ohio, Wyoming to Nevada (through this part) *shape: categorical; highest to lowest: Arizona, Ohio, Wyoming, state other than Massachusetts, Missouri, Arizona, Nevada, Ohio, Wyoming, Massachusetts, Missouri …*	1% to 1%	log-odds -0.26, OR 0.77; range across patients -0.26 to -0.26	3508, 77	train; test disagrees (AUC gain -0.0003 [-0.001, 0.001], n=1000)	yes	slightly; on the training data only
C141	direction: state: state other than Massachusetts, Missouri, Arizona, Nevada, Ohio, Wyoming to Missouri (through this part) *shape: categorical; highest to lowest: Arizona, Ohio, Wyoming, state other than Massachusetts, Missouri, Arizona, Nevada, Ohio, Wyoming, Massachusetts, Missouri …*	1% to 1%	log-odds -0.04, OR 0.97; range across patients -0.04 to -0.04	3508, 80	train; test disagrees (AUC gain -0.0003 [-0.001, 0.001], n=1000)	yes	slightly; on the training data only
C142	direction: state: state other than Massachusetts, Missouri, Arizona, Nevada, Ohio, Wyoming to Massachusetts (through this part) *shape: categorical; highest to lowest: Arizona, Ohio, Wyoming, state other than Massachusetts, Missouri, Arizona, Nevada, Ohio, Wyoming, Massachusetts, Missouri …*	1% to 1%	log-odds -0.001, OR 1.00; range across patients -0.001 to -0.001	3508, 86	train; test disagrees (AUC gain -0.0003 [-0.001, 0.001], n=1000)	yes	slightly; on the training data only
C143	structure: its formula reproduces it on 4000 training patients (largest difference 2.22045e-16)		log-odds 2e-16, OR n/a	4000	train	yes	

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C145	direction: total daytime calls: 75 calls to 125 calls (through this part) *shape: flat; 100% of patients same; verified; risk saturates near 0%*	1% to 1%	log-odds -0.01, OR 0.99; range across patients -0.01 to -0.01	4000	train; test unclear (AUC gain 4e-05 [-0.0002, 0.0003], n=1000)	yes	slightly; on the training data only
C146	interaction: effect of total daytime calls (75 calls to 125 calls) by state: Tennessee minus state other than Tennessee; stronger in state other than Tennessee *shape: steady; 100% of patients same; verified; risk saturates near 0%*	Tennessee: 2% to 3%; state other than Tennessee: 1% to 1%	log-odds difference -0.003, OR 1.00	68, 3932	train; test agrees (AUC gain -0.0006 [-0.001, -7e-05], n=1000); group difference on test -0.003 [-0.003, -0.003]	no	slightly
C147	threshold: total daytime calls: fold at about 114 calls (stops) *shape: flat; 100% of patients same; verified; risk saturates near 0%*	1% to 1% (typical patient with a fold)	fold 113 calls (IQR 113 to 113); past it log-odds -0.01, OR 0.99; range across patients -0.01 to -0.01	3932, 2926	train; test unclear (AUC gain 4e-05 [-0.0002, 0.0003], n=1000)	no	slightly; on the training data only
C148	subgroup: who this part acts on: 2994 of 4000 patients	1% to 1% (typical patient this part acts on (risk without the part, then with it))	share 0.75; log-odds 0.01, OR 1.01; range across patients 0.001 to 0.01	2994, 1006	train; test disagrees (AUC gain -0.0005 [-0.001, 9e-06], n=1000)	no	slightly; on the training data only
C149	structure: its formula reproduces it on 4000 training patients (largest difference 5.55112e-17)		log-odds 6e-17, OR n/a	4000	train	yes	
C150	direction: state: state other than Arkansas, New Mexico, Utah to Arkansas (through this part) *shape: categorical; highest to lowest: Arkansas, Utah, New Mexico, state other than Arkansas, New Mexico, Utah*	1% to 1%	log-odds 0.05, OR 1.06; range across patients 0.03 to 0.08	3761, 78	train; test disagrees (AUC gain -5e-05 [-0.0002, 0.0001], n=1000)	yes	slightly; on the training data only
C151	direction: total international charge: 1.84 US dollars to 3.70 US dollars (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 1% (typical patient this part acts on)	log-odds -0.001, OR 1.00; range across patients 0.00 to 0.00	4000	train; test unclear (AUC gain 2e-05 [-6e-05, 0.0001], n=1000)	yes	slightly; on the training data only
C152	direction: customer-service calls: 0 calls to 3 calls (through this part) *shape: flat; 100% of patients same; verified; risk saturates near 0%*	1% to 1% (typical patient this part acts on)	log-odds -0.0003, OR 1.00; range across patients 0.00 to 0.00	4000	train; test unclear (AUC gain 0.000 [0.000, 0.000], n=1000)	yes	slightly; on the training data only
C153	direction: state: state other than Arkansas, New Mexico, Utah to Utah (through this part) *shape: categorical; highest to lowest: Arkansas, Utah, New Mexico, state other than Arkansas, New Mexico, Utah*	1% to 1%	log-odds 0.009, OR 1.01; range across patients 0.00 to 0.03	3761, 89	train; test disagrees (AUC gain -5e-05 [-0.0002, 0.0001], n=1000)	yes	slightly; on the training data only
C154	interaction: effect of customer-service calls (0 calls to 3 calls) by state: Arkansas minus state other than Arkansas, New Mexico, Utah; stronger in Arkansas *shape: accelerating; 82% of patients same; fallback; risk saturates near 0%*	Arkansas: 2% to 4%; state other than Arkansas, New Mexico, Utah: 1% to 3%	log-odds difference 0.28, OR 1.32	78, 3761	train; test disagrees (AUC gain -5e-05 [-0.0002, 0.0001], n=1000); group difference on test -0.27 [-0.85, 0.37]	no	slightly; on the training data only
C155	interaction: effect of total international charge (1.84 US dollars to 3.70 US dollars) by state: Arkansas minus state other than Arkansas, New Mexico, Utah; stronger in state other than Arkansas, New Mexico, Utah *shape: accelerating; 91% of patients same; verified; risk saturates near 0%*	Arkansas: 1% to 3%; state other than Arkansas, New Mexico, Utah: 1% to 2%	log-odds difference -0.07, OR 0.93	78, 3761	train; test disagrees (AUC gain -5e-05 [-0.0002, 0.0001], n=1000); group difference on test 0.01 [-0.12, 0.14]	no	slightly; on the training data only
C156	interaction: effect of customer-service calls (0 calls to 3 calls) by total international charge: total international charge above 2.78 US dollars minus total international charge at or below 2.78 US dollars; stronger in total international charge at or below 2.78 US dollars *shape: accelerating; 78% of patients same; fallback; risk saturates near 0%*	total international charge above 2.78 US dollars: 2% to 3%; total international charge at or below 2.78 US dollars: 1% to 3%	log-odds difference -0.06, OR 0.94	1999, 2001	train; test unclear (AUC gain 2e-05 [-6e-05, 0.0001], n=1000); group difference on test -0.12 [-0.26, 0.03]	no	slightly; on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C157	threshold: customer-service calls: fold at about 5 calls (stops) *shape: flat; 100% of patients same; verified; risk saturates near 0%*	2% to 2% (typical patient with a fold)	fold 4.49 calls (IQR 3.72 to 5.40); past it log-odds -0.01, OR 0.99; range across patients -0.02 to -0.004	66, 63	train; test unclear (AUC gain 0.000 [0.000, 0.000], n=1000)	no	slightly; on the training data only
C158	threshold: total international charge: fold at about 0.43 US dollars (stops) *shape: flat; 97% of patients same; verified; risk saturates near 0%*	1% to 1% (typical patient with a fold)	fold 0.43 US dollars (IQR 0.34 to 0.43); past it log-odds -0.002, OR 1.00; range across patients 0.00 to 0.00	3890, 153	train; test unclear (AUC gain 2e-05 [-6e-05, 0.0001], n=1000)	no	slightly; on the training data only
C159	subgroup: who this part acts on: 153 of 4000 patients	1% to 1% (typical patient this part acts on (risk without the part, then with it))	share 0.04; log-odds 0.04, OR 1.04; range across patients 0.005 to 0.07	153, 3847	train; test disagrees (AUC gain -2e-05 [-0.0002, 0.0002], n=1000)	no	slightly; on the training data only
C160	null: this part barely changes predictions (switched off)		max change 0.020 (max |change in predicted risk|)	4000	train	no	
C161	structure: its formula reproduces it on 4000 training patients (largest difference 8.88178e-16)		log-odds 9e-16, OR n/a	4000	train	yes	
C162	boundary: boundary		log-odds n/a, OR n/a	4000	train	no	
C163	boundary: boundary		log-odds n/a, OR n/a	4000	train	no	
C164	boundary: boundary		log-odds 1.00, OR n/a	4000	train	yes	

Not supported by the model.

- C002: direction claim about total daytime minutes in the whole model: not supported by the model (17% of patients have the opposite sign (allowed: 10%)).
- C017: direction claim about total evening charge in the whole model: not supported by the model (52% of patients have the opposite sign (allowed: 10%)).
- C041: direction claim about state in the whole model: not supported by the model (45% of patients have the opposite sign (allowed: 10%)).
- C044: direction claim about state in the whole model: not supported by the model (92% of patients have the opposite sign (allowed: 10%)).
- C146: interaction claim about total daytime calls and state in part ann_97: not supported by the model (the group difference is below 0.1 log-odds).
- C147: threshold claim about total daytime calls in part ann_97: not supported by the model (the fold changes the curve by only 0.014 log-odds compared with a straight line (needed 0.1); the effect past the fold is too small or not consistent across patients).
- C148: subgroup claim about state and total daytime calls in part ann_97: not supported by the model (its effect is negligible).
- C154: interaction claim about number of calls to customer service and state in part ann_92: not supported by the model (the part's own counterfactual difference has the opposite sign).
- C155: interaction claim about total international charge and state in part ann_92: not supported by the model (the group difference is below 0.1 log-odds).
- C156: interaction claim about number of calls to customer service and total international charge in part ann_92: not supported by the model (the group difference is below 0.1 log-odds; the part's own counterfactual difference has the opposite sign).
- C157: threshold claim about number of calls to customer service in part ann_92: not supported by the model (the fold changes the curve by only 0.015 log-odds compared with a straight line (needed 0.1); the effect past the fold is too small or not consistent across patients).
- C158: threshold claim about total international charge in part ann_92: not supported by the model (the effect past the fold is too small or not consistent across patients).
- C159: subgroup claim about state and total international charge in part ann_92: not supported by the model (its effect is negligible).
- C160: null claim about state and total international charge in part ann_92: not supported by the model (predictions move by up to 0.020).

- C162: boundary claim about total daytime minutes and number of calls to customer service in part ann_101: not supported by the model (no switch-on line could be traced on the first map to check against the data; no switch-on boundary in the observed range).
- C163: boundary claim about total daytime minutes and number of calls to customer service in part ann_91: not supported by the model (no switch-on line could be traced on the first map to check against the data; no switch-on boundary in the observed range).

1.2 Coverage

Coverage is computed by the current code (the function behind the `get_coverage` tool) under these conventions. *Candidate nodes* are all nodes of the explained model state except the output nodes. A candidate node is *covered* when some annotation of the hierarchy (a leaf, or a composition together with its descendants) contains it and all its enabled outgoing edges. *Exact Null* parts, which have no effect on any output for any input, count as covered: nodes with no path to an output over enabled, non-zero-weight edges (disconnected inputs and dead ends), and edges that have zero weight or touch such a node. Compositional coverage is the fraction of required compositions present: every annotation except a single global root containing every output must have a parent; a parentless, childless annotation made only of Null nodes is exempt.

- **Null, disconnected input (24):** `-10` (state:17), `-11` (state:18), `-15` (state:21), `-16` (state:22), `-17` (state:23), `-19` (state:25), `-26` (state:31), `-29` (state:34), `-3` (state:10), `-34` (state:39), `-36` (state:40), `-39` (state:43), `-49` (state:7), `-5` (state:12), `-50` (state:8), `-51` (state:9), `-6` (state:13), `-61` (totalDayCharge), `-63` (totalEveCalls), `-67` (totalNightCharge), `-69` (totalIntlCalls), `-7` (state:14), `-8` (state:15), `-9` (state:16).
- **Uncovered nodes:** none.
- **Uncovered edges:** none.
- **Missing compositions:** none.

1.3 Annotation hierarchy

The hierarchy has 1 root; the global root is churn_logit. Tags (A1, A2, . . .) are used in the figures.

▷ A1: churn_logit (composition, section 1.6.1)
- ▷ A2: intl_state_unit (leaf, section 1.6.2)
- ▷ A3: eve_minutes_hinge (leaf, section 1.6.3)
- ▷ A4: voicemail_chain (leaf, section 1.6.4)
- ▷ A5: state_offsets (leaf, section 1.6.5)
- ▷ A6: day_calls_state42 (leaf, section 1.6.6)
- ▷ A7: linear_backbone (leaf, section 1.6.7)
- ▷ A8: service_call_nonlinear (composition, section 1.6.8)
  - ▷ A9: csc_usage_gate (leaf, section 1.6.9)
  - ▷ A10: csc_hinge (leaf, section 1.6.10)

1.4 Network figures

The explained model state has more than 60 non-Null nodes, so the collapsed views are shown; the full view is in `churn100-s2-c1-fig-full.tex`.

1.5 Refactoring operations

The explained model state is the phenotype after these operations, replayed in order. All are function-preserving; the largest change in the model output over the 4000 training rows is 6.66×10^{-16} (tolerance 1.00×10^{-9}).

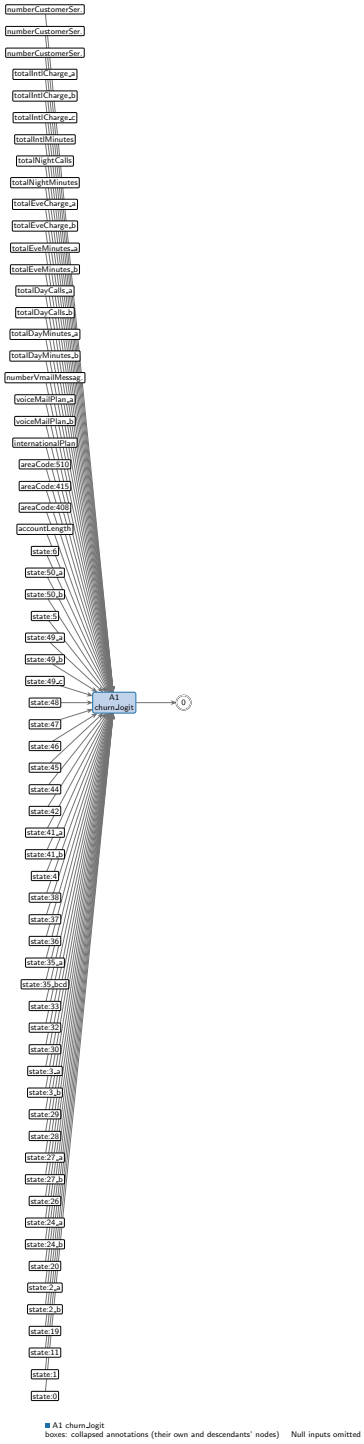


Figure 1: churn100-s2-c1: collapsed view at hierarchy level 0 (fully recomposed).

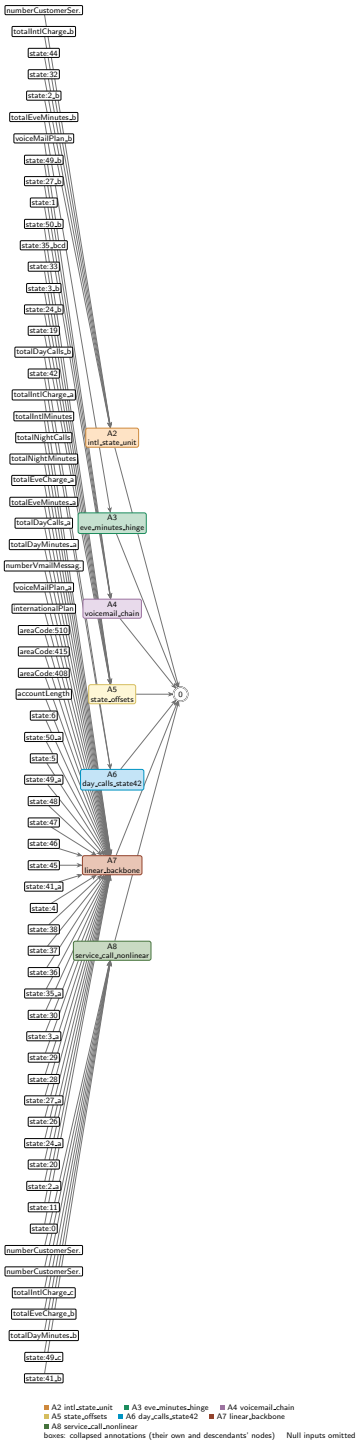


Figure 2: churn100-s2-c1: collapsed view at hierarchy level 1.

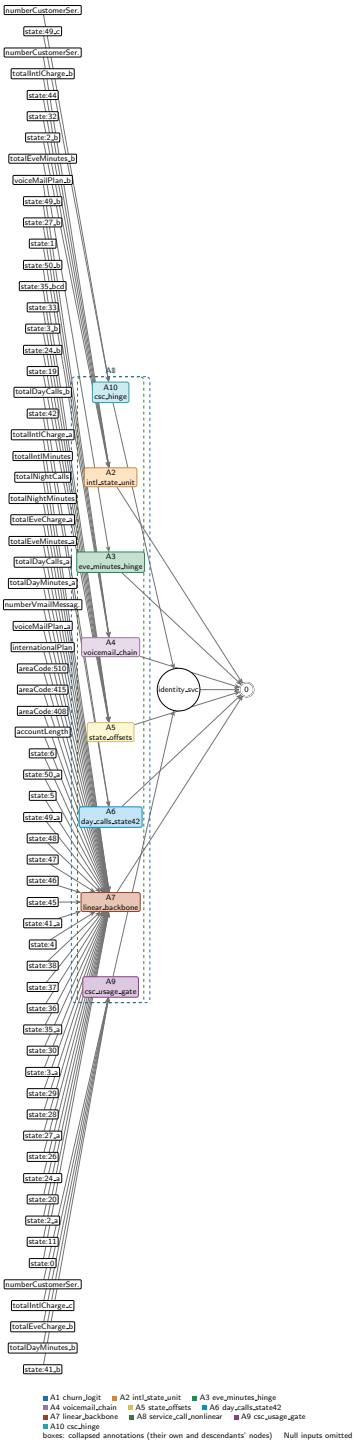


Figure 3: churn100-s2-c1: collapsed view at hierarchy level 2.

- #71: split 2627 into 2627_a, 2627_b.
- #72: split 9555 into 9555_a, 9555_b.
- #73: split -71 into -71_a, -71_b, -71_c, -71_d, -71_e, -71_f, -71_g, -71_h.
- #74: split -70 into -70_a, -70_b, -70_c.
- #75: split -64 into -64_a, -64_b.
- #76: split -62 into -62_a, -62_b.
- #77: split -60 into -60_a, -60_b.
- #78: split -59 into -59_a, -59_b.
- #79: split -57 into -57_a, -57_b.
- #80: split -47 into -47_a, -47_b.
- #81: split -45 into -45_a, -45_b, -45_c.
- #82: split -37 into -37_a, -37_b.
- #83: split -30 into -30_a, -30_b, -30_c, -30_d.
- #84: split -24 into -24_a, -24_b.
- #85: split -21 into -21_a, -21_b.
- #86: split -18 into -18_a, -18_b.
- #87: split -13 into -13_a, -13_b.
- #88: consolidate -71_a, -71_b, -71_c, -71_f, -71_h.
- #89: consolidate -71_e, -71_g.
- #90: consolidate -30_b, -30_c, -30_d.
- #98: identity node identity_lin before 0 (40 connection(s)).
- #100: identity node identity_svc before 0 (3 connection(s)).

1.6 Annotations

1.6.1 A1: churn_logit

Kind: Composition annotation (depth 0); children A2 (section 1.6.2), A3 (section 1.6.3), A4 (section 1.6.4), A5 (section 1.6.5), A6 (section 1.6.6), A7 (section 1.6.7), A8 (section 1.6.8).

- **Entry nodes:** -69 (totalIntlCalls), -68 (totalIntlMinutes), -67 (totalNightCharge), -66 (totalNightCalls), -65 (totalNightMinutes), -63 (totalEveCalls), -61 (totalDayCharge), -58 (numberVmailMessages), -56 (internationalPlan), -55 (areaCode:510), -54 (areaCode:415), -53 (areaCode:408), -52 (accountLength), -51 (state:9), -50 (state:8), -49 (state:7), -48 (state:6), -46 (state:5), -44 (state:48), -43 (state:47), -42 (state:46), -41 (state:45), -40 (state:44), -39 (state:43), -38 (state:42), -36 (state:40), -35 (state:4), -34 (state:39), -33 (state:38), -32 (state:37), -31 (state:36), -29 (state:34), -28 (state:33), -27 (state:32), -26 (state:31), -25 (state:30), -23 (state:29), -22 (state:28), -20 (state:26), -19 (state:25), -17 (state:23), -16 (state:22), -15 (state:21), -14 (state:20), -12 (state:19), -11 (state:18), -10 (state:17), -9 (state:16), -8 (state:15), -7 (state:14), -6 (state:13), -5 (state:12), -4 (state:11), -3 (state:10), -2 (state:1), -1 (state:0), -71_d (numberCustomerServiceCalls_d), -70_a (totalIntlCharge_a), -70_b (totalIntlCharge_b), -70_c (totalIntlCharge_c), ... (29 more; see the JSON record).

- **Exit nodes:** 0.
- **Internal nodes:** none.
- **Edges (weight):** 252→0 (0.0205), 476→0 (-0.55), 1133→0 (-0.95), 3861→0 (-0.15), 4051→0 (-0.994), 8896→0 (0.0326), 10249→0 (0.04), 11427→0 (0.0502), 13827→0 (0.432), 14942→0 (0.0416), identity_lin→0 (1), identity_svc→0 (1).
- **With its descendants:** 116 nodes.
- **Covers** (coverage, own and as part of the hierarchy): 115 candidate nodes.

Hypothesis. Root: logit P(churn) = -1.05 + linear_backbone + service_call_nonlinear (identity_svc = 1.60*807 + 1.81*1264 + 1.48*7208) - 0.95*1133 - 0.99*4051 (voicemail_chain) - 0.55*476 (eve_minutes_hinge) + state_offsets + 0.02*252 + 0.05*11427, purely additive in the logit; the sigmoid is the only coupling. 24 inputs are unconnected (total intl calls, night charge, evening calls, day charge and 20 state indicators) and have zero effect. Train AUC 0.901, accuracy 0.901. Performance sources (train): all children replaced by least-squares linear fits gives AUC 0.837 (an L2 logistic regression refit on all 71 inputs gets 0.843); restoring csc_usage_gate alone gives 0.898, csc_hinge alone 0.851, eve_minutes_hinge alone 0.834, the others 0.837 (exactly or nearly linear). So the gain over linear comes from the service-call x low-usage gate. Rare-category switches: state:11/46/5 offsets of -2.24/-1.89/-1.46 logit on 73-80 customers (empirical -1.35/-0.91/-0.51); state:4 +1.35 (41 rows, empirical +0.80); state:47 +1.16 (72 rows, +0.92); state:49 -0.42 offset, down to -2.87 for 12 customers with 4+ calls (turns off the 4+-calls hinge); state:35 +0.40 (empirical +0.06).

Closed form (children as named functions):

$$
\begin{aligned}
y_0 &= \sigma(z_0) \\
z_0 &= -1.05 + 1.0\,\text{linear_backbone}\big(x_{state:0}, x_{state:11}, \\
&\quad x_{state:20}, x_{state:26}, x_{state:28}, x_{state:29}, x_{state:30}, x_{state:36}, \\
&\quad x_{state:37}, x_{state:38}, x_{state:4}, x_{state:45}, x_{state:46}, x_{state:47}, \\
&\quad x_{state:48}, x_{state:5}, x_{state:6}, x_{accountLength}, x_{areaCode:408}, x_{areaCode:415}, \\
&\quad x_{areaCode:510}, x_{internationalPlan}, x_{numberVmailMessages}, x_{totalNightMinutes}, \\
&\quad x_{totalNightCalls}, x_{totalIntlMinutes}, x_{totalIntlCharge_a}, x_{totalEveCharge_a}, \\
&\quad x_{totalEveMinutes_a}, x_{totalDayCalls_a}, x_{totalDayMinutes_a}, x_{voiceMailPlan_a}, \\
&\quad x_{state:50_a}, x_{state:49_a}, x_{state:41_a}, x_{state:35_a}, x_{state:3_a}, \\
&\quad x_{state:27_a}, x_{state:24_a}, x_{state:2_a}\big) \\
&\quad + 1.0\,\text{service_call_nonlinear}\big(x_{numberCustomerServiceCalls_abcfh}, \\
&\quad x_{totalEveCharge_b}, x_{state:41_b}, x_{totalDayMinutes_b}, x_{totalIntlCharge_c}, \\
&\quad x_{numberCustomerServiceCalls_d}, x_{state:49_c}\big) \\
&\quad + 0.04\,\text{state_offsets_1}\big(x_{state:19}, x_{state:3_b}, x_{state:35_bcd}, \\
&\quad x_{state:33}, x_{state:24_b}, x_{state:50_b}\big) \\
&\quad + 0.0416\,\text{state_offsets_2}\big(x_{state:19}, x_{state:3_b}, \\
&\quad x_{state:35_bcd}, x_{state:33}, x_{state:24_b}, x_{state:50_b}\big) \\
&\quad + 0.0326\,\text{state_offsets_0}\big(x_{state:19}, x_{state:3_b}, \\
&\quad x_{state:35_bcd}, x_{state:33}, x_{state:24_b}, x_{state:50_b}\big) \\
&\quad + 0.0205\,\text{intl_state_unit}\big(x_{numberCustomerServiceCalls_eg}, \\
&\quad x_{state:32}, x_{state:2_b}, x_{state:44}, x_{totalIntlCharge_b}\big) \\
&\quad + 0.0502\,\text{day_calls_state42}\big(x_{state:42}, x_{totalDayCalls_b}\big) \\
&\quad + 0.432\,\text{state_offsets_3}\big(x_{state:19}, x_{state:3_b}, \\
&\quad x_{state:35_bcd}, x_{state:33}, x_{state:24_b}, x_{state:50_b}\big) \\
&\quad - 0.55\,\text{eve_minutes_hinge}\big(x_{totalEveMinutes_b}\big)
\end{aligned}
$$

$$
\begin{aligned}
&- 0.15\,\text{state_offsets_4}\big(x_{state:19}, x_{state:3_b}, x_{state:35_bcd},\\
&x_{state:33}, x_{state:24_b}, x_{state:50_b}\big)\\
&- 0.95\,\text{voicemail_chain_0}\big(x_{voiceMailPlan_b}, x_{state:1},\\
&x_{state:27_b}, x_{state:49_b}\big)\\
&- 0.994\,\text{voicemail_chain_1}\big(x_{voiceMailPlan_b}, x_{state:1},\\
&x_{state:27_b}, x_{state:49_b}\big)
\end{aligned}
$$

The fully expanded form is in the JSON record. where y_0 is node `0`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.00582, and the exact function by at most 6.66×10^{-16}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. Root: logit P(churn) = -1.05 + linear_backbone + service_call_nonlinear (1.60*807 + 1.81*1264 + 1.48*7208) - 0.95*1133 - 0.99*4051 (voicemail_chain) - 0.55*476 (eve_minutes_hinge) + state_offsets + 0.02*252 + 0.05*11427, purely additive in the logit; the sigmoid is the only coupling. 24 inputs are unconnected (total intl calls, night charge, evening calls, day charge and 20 state indicators) and h

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1.6.2 A2: intl_state_unit

Kind: Leaf annotation (depth 1); parent A1 (section 1.6.1).

- **Entry nodes:** `-71_eg` (numberCustomerServiceCalls_eg), `-27` (state:32), `-13_b` (state:2_b), `-40` (state:44), `-70_b` (totalIntlCharge_b).
- **Exit nodes:** `252`.
- **Internal nodes:** `9555_a`, `2627_a`, `161`.
- **Edges (weight):** `-27`→`161` (-0.0113), `-40`→`252` (0.437), `161`→`252` (0.791), `2627_a`→`252` (-0.378), `9555_a`→`2627_a` (0.325), `-70_b`→`252` (-0.869), `-13_b`→`252` (0.76), `-71_eg`→`2627_a` (0.278), `-71_eg`→`9555_a` (0.855).
- **Covers** (coverage, own and as part of the hierarchy): 8 candidate nodes.

Hypothesis. Near-dead unit 252 = relu(-2.22 + 0.76 state:2 + 0.44 state:44 - 0.87 intlCharge + 0.79*161 - 0.38 g(csc)), with 161 = relu(-0.14 - 0.011 state:32) identically 0 on train (dead). 252 is active on 153 of 4000 training rows (78 in state:2, 55 in state:44, mostly with low international charge); its logit contribution is at most +0.10 (weight 0.02). Hypothesis: a negligible state-specific low-international-charge offset. Evidence: mean ablation and linear replacement leave train AUC at 0.9009; restoring it onto the all-linear model changes AUC by -0.0001.

Closed form:

$$
\begin{aligned}
y_0 = \text{relu}\big(&-2.22 + 0.76 x_{state:2_b}\\
&+ 0.791\,\text{relu}\big(-0.14 - 0.0113 x_{state:32}\big) + 0.437 x_{state:44}\\
&- 0.378\,\text{relu}\big(1.01 + 0.325\,\text{relu}\big(-0.0966\\
&+ 0.855 x_{numberCustomerServiceCalls_eg}\big)\\
&+ 0.278 x_{numberCustomerServiceCalls_eg}\big) - 0.869 x_{totalIntlCharge_b}\big)
\end{aligned}
$$

where y_0 is node `252`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.00409, and the exact function by at most 8.88×10^{-16}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. Near-dead unit 252 = relu(-2.22 + 0.76 state:2 + 0.44 state:44 - 0.87 intlCharge + 0.79*161 - 0.38 g(csc)), with 161 = relu(-0.14 - 0.011 state:32) identically 0 on train (dead). 252 is active on 153 of 4000 training rows (78 in state:2, 55 in state:44, mostly with low international charge); its logit contribution is at most +0.10 (weight 0.02). Hypothesis: a negligible state-specific low-internat

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

1.6.3 A3: eve_minutes_hinge

Kind: Leaf annotation (depth 1); parent A1 (section 1.6.1).

- **Entry nodes:** `-62_b` (totalEveMinutes_b).
- **Exit nodes:** `476`.
- **Internal nodes:** none.
- **Edges (weight):** `-62_b`→`476` (1.07).
- **Covers** (coverage, own and as part of the hierarchy): 1 candidate node.

Hypothesis. 476 = relu(1.07 eveMinutes - 0.81), logit weight -0.55. Hypothesis: above about 239 evening minutes (885 of 4000 training customers) this hinge flattens the direct evening-minutes slope (+0.51 per SD direct, -0.59 per SD through the hinge), so evening minutes stop adding risk above about 239 minutes (evening charge, +0.45 per SD, still adds). Evidence: line plot of 476 on train from 0 to 2.7; mean ablation AUC 0.9002, linear replacement 0.9005; restoring it alone onto the all-linear model lowers AUC (0.834 vs 0.837): a small fit-to-train detail.

Closed form:

$$y_0 = \mathrm{relu}\big(-0.805 + 1.07 x_{totalEveMinutes_b}\big)$$

where y_0 is node `476`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.00493, and the exact function by at most 0.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. 476 = relu(1.07 eveMinutes - 0.81), logit weight -0.55. Hypothesis: above about 239 evening minutes (885 of 4000 training customers) this hinge flattens the direct evening-minutes slope (+0.51 per SD direct, -0.59 per SD through the hinge), so evening minutes stop adding risk above about 239 minutes (evening charge, +0.45 per SD, still adds). Evidence: line plot of 476 on train from 0 to 2.7; mean

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1.6.4 A4: voicemail_chain

Kind: Leaf annotation (depth 1); parent A1 (section 1.6.1).

- **Entry nodes:** `-57_b` (voiceMailPlan_b), `-2` (state:1), `-21_b` (state:27_b), `-45_b` (state:49_b).
- **Exit nodes:** `1133`, `4051`.
- **Internal nodes:** `5577`.

- **Edges (weight):** `-2`→`5577` (0.255), `1133`→`4051` (1.15), `5577`→`1133` (0.137), `-57_b`→`5577` (0.991), `-45_b`→`4051` (0.0611), `-21_b`→`1133` (0.0189).
- **Covers** (coverage, own and as part of the hierarchy): 5 candidate nodes.

Hypothesis. 5577 = relu(0.99 + 0.99 voiceMailPlan + 0.26 state:1), 1133 = relu(0.68 + 0.14*5577 + 0.02 state:27), 4051 = relu(2.09 + 1.15*1133 + 0.06 state:49). All three are always active on train (minimum pre-activations 0.35, 0.73, 2.91), so the chain is exactly affine in its binary inputs (least-squares R2 = 1.0). Hypothesis: a linear voicemail-plan term (net logit -0.95*1133 - 0.99*4051 = about -0.28 per SD of voicemail plan on top of the direct -0.90) plus small state:1/27/49 offsets. Evidence: exact linear fit; linear replacement AUC 0.9009 (unchanged).

Closed form:

$$
\begin{aligned}
y_0 &= \operatorname{relu}\big(0.68 + 0.137\operatorname{relu}\big(0.991 \\
&\qquad + 0.255x_{state:1} + 0.991x_{voiceMailPlan_b}\big) + 0.0189x_{state:27_b}\big) \\
y_1 &= \operatorname{relu}\big(2.09 + 1.15\operatorname{relu}\big(0.68 \\
&\qquad + 0.137\operatorname{relu}\big(0.991 + 0.255x_{state:1} + 0.991x_{voiceMailPlan_b}\big) \\
&\qquad + 0.0189x_{state:27_b}\big) + 0.0611x_{state:49_b}\big)
\end{aligned}
$$

where y_0 is node `1133`; y_1 is node `4051`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.01, and the exact function by at most 4.44×10^{-16}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. 5577 = relu(0.99 + 0.99 voiceMailPlan + 0.26 state:1), 1133 = relu(0.68 + 0.14*5577 + 0.02 state:27), 4051 = relu(2.09 + 1.15*1133 + 0.06 state:49). All three are always active on train (minimum pre-activations 0.35, 0.73, 2.91), so the chain is exactly affine in its binary inputs (least-squares R2 = 1.0). Hypothesis: a linear voicemail-plan term (net logit -0.95*1133 - 0.99*4051 = about -0.29 per

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

1.6.5 A5: state_offsets

Kind: Leaf annotation (depth 1); parent A1 (section 1.6.1).

- **Entry nodes:** `-12` (state:19), `-24_b` (state:3_b), `-30_bcd` (state:35_bcd), `-28` (state:33), `-18_b` (state:24_b), `-47_b` (state:50_b).
- **Exit nodes:** `8896`, `10249`, `14942`, `13827`, `3861`.
- **Internal nodes:** `11951`, `6730`.
- **Edges (weight):** `-12`→`8896` (-1), `-28`→`6730` (-0.418), `11951`→`6730` (-9.39×10^{-4}), `6730`→`13827` (1.01), `-47_b`→`3861` (-0.771), `-24_b`→`10249` (0.998), `-18_b`→`6730` (-0.0113), `-30_bcd`→`6730` (0.00244), `-30_bcd`→`11951` (1), `-30_bcd`→`14942` (1).
- **Covers** (coverage, own and as part of the hierarchy): 8 candidate nodes.

Hypothesis. Units 8896 (state:19), 10249 (state:3), 11951 and 14942 (state:35), 6730 -> 13827 (state:33, state:24, state:35), 3861 (state:50, identically 0 on train). Each depends only on state one-hot indicators; since the states are mutually exclusive, every such function is exactly an offset per state (least-squares R2 = 1.0 for every exit). Hypothesis: a per-state logit offset table; the largest

offsets here are state:33 about -0.25 (switches off 6730 -> 13827), state:35 +0.21 (14942), state:3 +0.30 (10249). Evidence: linear replacement AUC 0.9009 (unchanged); mean ablation 0.9006.

Closed form:

$$
\begin{aligned}
y_0 &= \mathrm{relu}\big(-0.106 - 1.0 x_{state:19}\big) \\
y_1 &= \mathrm{relu}\big(0.172 + 0.998 x_{state:3_b}\big) \\
y_2 &= \mathrm{relu}\big(-1.74 + 1.0 x_{state:35_bcd}\big) \\
y_3 &= \mathrm{relu}\big(-0.737 + 1.01\,\mathrm{relu}\big(1.26 \\
&\qquad + 0.00244 x_{state:35_bcd} - 0.0113 x_{state:24_b} - 0.000939\,\mathrm{relu}\big(-1.09 \\
&\qquad + 1.0 x_{state:35_bcd}\big) - 0.418 x_{state:33}\big)\big) \\
y_4 &= \mathrm{relu}\big(-0.431 - 0.771 x_{state:50_b}\big)
\end{aligned}
$$

where y_0 is node `8896`; y_1 is node `10249`; y_2 is node `14942`; y_3 is node `13827`; y_4 is node `3861`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.0312, and the exact function by at most 0.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. Units 8896 (state:19), 10249 (state:3), 11951 and 14942 (state:35), 6730 -> 13827 (state:33, state:24, state:35), 3861 (state:50, identically 0 on train). Each depends only on state one-hot indicators; since the states are mutually exclusive, every such function is exactly an offset per state (least-squares R2 = 1.0 for every exit). Hypothesis: a per-state logit offset table; the largest offsets h

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

1.6.6 A6: day_calls_state42

Kind: Leaf annotation (depth 1); parent A1 (section 1.6.1).

- **Entry nodes:** `-38` (state:42), `-60_b` (totalDayCalls_b).
- **Exit nodes:** `11427`.
- **Internal nodes:** none.
- **Edges (weight):** `-38`→`11427` (0.995), `-60_b`→`11427` (-0.105).
- **Covers** (coverage, own and as part of the hierarchy): 2 candidate nodes.

Hypothesis. 11427 = relu(0.20 + 1.00 state:42 - 0.11 dayCalls), logit weight +0.05. Hypothesis: a tiny hinge that is on for up to about 110 day calls and switches off above about 120 (on for 100% of rows below 110 day calls, 44% at 110-119, 2% at 120+), plus a state:42 offset; contribution at most +0.40 logit. Evidence: linear fit R2 0.999; mean ablation and linear replacement leave AUC at 0.9009.

Closed form:

$$
y_0 = \mathrm{relu}\big(0.198 + 0.995 x_{state:42} - 0.105 x_{totalDayCalls_b}\big)
$$

where y_0 is node `11427`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.0035, and the exact function by at most 0.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).

- EMPIRICAL, `ablation_study`; data split: train. 11427 = relu(0.20 + 1.00 state:42 - 0.11 dayCalls), logit weight +0.05. Hypothesis: a tiny hinge that is on for up to about 110 day calls and switches off above about 120 (on for 100% of rows below 110 day calls, 44% at 110-119, 2% at 120+), plus a state:42 offset; contribution at most +0.40 logit. Evidence: linear fit R2 0.999; mean ablation and linear replacement leave AUC at 0.9009.

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

1.6.7 A7: linear_backbone

Kind: Leaf annotation (depth 1); parent A1 (section 1.6.1).

- **Entry nodes:** `-1` (state:0), `-4` (state:11), `-14` (state:20), `-20` (state:26), `-22` (state:28), `-23` (state:29), `-25` (state:30), `-31` (state:36), `-32` (state:37), `-33` (state:38), `-35` (state:4), `-41` (state:45), `-42` (state:46), `-43` (state:47), `-44` (state:48), `-46` (state:5), `-48` (state:6), `-52` (accountLength), `-53` (areaCode:408), `-54` (areaCode:415), `-55` (areaCode:510), `-56` (internationalPlan), `-58` (numberVmailMessages), `-65` (totalNightMinutes), `-66` (totalNightCalls), `-68` (totalIntlMinutes), `-70_a` (totalIntlCharge_a), `-64_a` (totalEveCharge_a), `-62_a` (totalEveMinutes_a), `-60_a` (totalDayCalls_a), `-59_a` (totalDayMinutes_a), `-57_a` (voiceMailPlan_a), `-47_a` (state:50_a), `-45_a` (state:49_a), `-37_a` (state:41_a), `-30_a` (state:35_a), `-24_a` (state:3_a), `-21_a` (state:27_a), `-18_a` (state:24_a), `-13_a` (state:2_a).
- **Exit nodes:** `identity_lin`.
- **Internal nodes:** none.
- **Edges (weight):** `-1`→`identity_lin` (-0.104), `-4`→`identity_lin` (-0.3), `-14`→`identity_lin` (0.0772), `-20`→`identity_lin` (0.0915), `-22`→`identity_lin` (-0.0124), `-23`→`identity_lin` (-0.0533), `-25`→`identity_lin` (-0.0496), `-31`→`identity_lin` (-0.0184), `-32`→`identity_lin` (0.0293), `-33`→`identity_lin` (-0.105), `-35`→`identity_lin` (0.135), `-41`→`identity_lin` (-0.115), `-42`→`identity_lin` (-0.265), `-43`→`identity_lin` (0.154), `-44`→`identity_lin` (-0.0899), `-46`→`identity_lin` (-0.201), `-48`→`identity_lin` (0.0346), `-52`→`identity_lin` (0.138), `-53`→`identity_lin` (-0.0272), `-54`→`identity_lin` (0.0454), `-55`→`identity_lin` (-0.0161), `-56`→`identity_lin` (0.756), `-58`→`identity_lin` (0.623), `-65`→`identity_lin` (0.307), `-66`→`identity_lin` (-0.00711), `-68`→`identity_lin` (-0.0175), `-70_a`→`identity_lin` (0.705), `-64_a`→`identity_lin` (0.45), `-62_a`→`identity_lin` (0.511), `-60_a`→`identity_lin` (0.127), `-59_a`→`identity_lin` (2.04), `-57_a`→`identity_lin` (-0.899), `-47_a`→`identity_lin` (-0.112), `-45_a`→`identity_lin` (-0.0115), `-37_a`→`identity_lin` (-0.101), `-30_a`→`identity_lin` (0.0276), `-24_a`→`identity_lin` (-0.0222), `-21_a`→`identity_lin` (0.0267), `-18_a`→`identity_lin` (-0.074), `-13_a`→`identity_lin` (-0.0117).
- **Covers** (coverage, own and as part of the hierarchy): 40 candidate nodes.

Hypothesis. Direct input-to-output edges collected through identity_lin (identity). Exactly linear in 40 inputs: day minutes +2.04 per SD, voicemail plan -0.90, international plan +0.76, international charge +0.71, voicemail messages +0.62, evening minutes +0.51, evening charge +0.45, night minutes +0.31, state:11 -0.30, state:46 -0.27, state:5 -0.20, state:47 +0.15, account length +0.14, state:4 +0.14, day calls +0.13, and smaller state/area-code terms. Hypothesis: the additive backbone; day minutes and voicemail/international plan dominate. Evidence: closed form is exact (identity); sensitivities match the weights.

Closed form:

$$\begin{aligned} y_0 = {} & 0.154x_{state:47} + 0.135x_{state:4} + 0.623x_{numberVmailMessages} \\ & + 0.756x_{internationalPlan} + 0.0267x_{state:27_a} + 0.127x_{totalDayCalls_a} \\ & + 2.04x_{totalDayMinutes_a} + 0.0346x_{state:6} + 0.307x_{totalNightMinutes} \\ & + 0.0772x_{state:20} + 0.0915x_{state:26} + 0.45x_{totalEveCharge_a} \end{aligned}$$

$$
\begin{aligned}
&+ 0.511x_{totalEveMinutes_a} + 0.138x_{accountLength} + 0.705x_{totalIntlCharge_a} \\
&+ 0.0454x_{areaCode:415} + 0.0276x_{state:35_a} + 0.0293x_{state:37} \\
&- 0.074x_{state:24_a} - 0.0272x_{areaCode:408} - 0.0115x_{state:49_a} \\
&- 0.0496x_{state:30} - 0.265x_{state:46} - 0.0184x_{state:36} - 0.0899x_{state:48} \\
&- 0.0124x_{state:28} - 0.201x_{state:5} - 0.112x_{state:50_a} - 0.0161x_{areaCode:510} \\
&- 0.0175x_{totalIntlMinutes} - 0.3x_{state:11} - 0.0223x_{state:3_a} \\
&- 0.0117x_{state:2_a} - 0.101x_{state:41_a} - 0.104x_{state:0} - 0.105x_{state:38} \\
&- 0.0533x_{state:29} - 0.899x_{voiceMailPlan_a} - 0.00711x_{totalNightCalls} \\
&- 0.115x_{state:45}
\end{aligned}
$$

where y_0 is node `identity_lin`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.0152, and the exact function by at most 2.66×10^{-15}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `empirical_viz`; data split: train. Direct input-to-output edges collected through identity_lin (identity). Exactly linear in 40 inputs: day minutes +2.04 per SD, voicemail plan -0.90, international plan +0.76, international charge +0.71, voicemail messages +0.62, evening minutes +0.51, evening charge +0.45, night minutes +0.31, state:11 -0.30, state:46 -0.27, state:5 -0.20, state:47 +0.15, account length +0.14, state:4 +0.14, day c

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1.6.8 A8: service_call_nonlinear

Kind: Composition annotation (depth 1); parent A1 (section 1.6.1); children A9 (section 1.6.9), A10 (section 1.6.10).

- **Entry nodes:** `-71_abcfh` (numberCustomerServiceCalls_abcfh), `-64_b` (totalEveCharge_b), `-37_b` (state:41_b), `-59_b` (totalDayMinutes_b), `-70_c` (totalIntlCharge_c), `-71_d` (numberCustomerServiceCalls_d), `-45_c` (state:49_c).
- **Exit nodes:** `identity_svc`.
- **Internal nodes:** none.
- **Edges (weight):** `807`→`identity_svc` (1.6), `1264`→`identity_svc` (1.8), `7208`→`identity_svc` (1.48).
- **With its descendants:** 16 nodes.
- **Covers** (coverage, own and as part of the hierarchy): 15 candidate nodes.

Hypothesis. Composition of csc_usage_gate and csc_hinge: their exits are summed by identity_svc = 1.60*807 + 1.81*1264 + 1.48*7208, which enters the logit with weight 1; they share only the customer-service-calls input (split copies). Hypothesis: all of the model's service-call effect is nonlinear (calls have no direct edge): risk jumps from 4 calls, and for frequent callers low day usage raises risk further (the gate). Evidence: mean-ablating both gives AUC 0.736, linear replacement of both 0.834 (all-linear model 0.837); the gate alone restores 0.898 of 0.901.

Closed form (children as named functions):

$$
\begin{aligned}
y_0 = 1.8\,\mathrm{csc_usage_gate_1}\big(&x_{numberCustomerServiceCalls_abcfh}, \\
&x_{totalEveCharge_b}, x_{state:41_b}, x_{totalDayMinutes_b}, x_{totalIntlCharge_c}\big)
\end{aligned}
$$

$$
\begin{aligned}
&+ 1.6\,\mathrm{csc_usage_gate_0}\big(x_{numberCustomerServiceCalls_abcfh},\\
&x_{totalEveCharge_b}, x_{state:41_b}, x_{totalDayMinutes_b}, x_{totalIntlCharge_c}\big)\\
&+ 1.48\,\mathrm{csc_hinge}\big(x_{numberCustomerServiceCalls_d}, x_{state:49_c}\big)
\end{aligned}
$$

Expanded over the children:

$$
\begin{aligned}
y_0 = 1.8\,\mathrm{relu}\big(&0.0759 + 0.363\,\mathrm{relu}\big(-1.29\\
&+ 0.622\,\mathrm{relu}\big(0.416 + 0.154x_{state:41_b}\\
&+ 0.342\,\mathrm{relu}\big(-0.649 + 0.987x_{numberCustomerServiceCalls_abcfh}\big)\\
&+ 0.36x_{numberCustomerServiceCalls_abcfh} - 1.0x_{totalEveCharge_b}\big)\\
&+ 0.0405x_{numberCustomerServiceCalls_abcfh} + 0.675\,\mathrm{relu}\big(1.01\\
&+ 0.325\,\mathrm{relu}\big(-0.0966 + 0.855x_{numberCustomerServiceCalls_abcfh}\big)\\
&+ 0.278x_{numberCustomerServiceCalls_abcfh}\big) - 1.4x_{totalDayMinutes_b}\big)\\
&- 0.326x_{totalIntlCharge_c}\big) + 1.6\,\mathrm{relu}\big(-1.29\\
&+ 0.622\,\mathrm{relu}\big(0.416 + 0.154x_{state:41_b}\\
&+ 0.342\,\mathrm{relu}\big(-0.649 + 0.987x_{numberCustomerServiceCalls_abcfh}\big)\\
&+ 0.36x_{numberCustomerServiceCalls_abcfh} - 1.0x_{totalEveCharge_b}\big)\\
&+ 0.0405x_{numberCustomerServiceCalls_abcfh} + 0.675\,\mathrm{relu}\big(1.01\\
&+ 0.325\,\mathrm{relu}\big(-0.0966 + 0.855x_{numberCustomerServiceCalls_abcfh}\big)\\
&+ 0.278x_{numberCustomerServiceCalls_abcfh}\big) - 1.4x_{totalDayMinutes_b}\big)\\
&+ 1.48\,\mathrm{relu}\big(-0.673 + 0.544\,\mathrm{relu}\big(0.00784\\
&+ 1.0x_{numberCustomerServiceCalls_d}\big) - 0.359x_{state:49_c}\big)
\end{aligned}
$$

where y_0 is node `identity_svc`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.0274, and the exact function by at most 3.55×10^{-15}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. Composition of csc_usage_gate and csc_hinge: their exits enter the logit additively (+1.60*807 + 1.81*1264 + 1.48*7208); they share only the customer-service-calls input (split copies). Hypothesis: all of the model's service-call effect is nonlinear (calls have no direct edge): risk jumps from 4 calls, and for frequent callers low day usage raises risk further (the gate). Evidence: mean-ablating b

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1.6.9 A9: csc_usage_gate

Kind: Leaf annotation (depth 2); parent A8 (section 1.6.8).

- **Entry nodes:** `-71_abcfh` (numberCustomerServiceCalls_abcfh), `-64_b` (totalEveCharge_b), `-37_b` (state:41_b), `-59_b` (totalDayMinutes_b), `-70_c` (totalIntlCharge_c).
- **Exit nodes:** `807`, `1264`.
- **Internal nodes:** `9555_b`, `2627_b`, `12926`, `2796`.
- **Edges (weight):** `2796`→`807` (0.622), `807`→`1264` (0.363), `12926`→`2796` (0.342), `2627_b`→`807` (0.675), `9555_b`→`2627_b` (0.325), `-70_c`→`1264` (-0.326), `-64_b`→`2796` (-1), `-59_b`→`807` (-1.4), `-37_b`→`2796` (0.154), `-71_abcfh`→`807` (0.0405), `-71_abcfh`→`2796` (0.36), `-71_abcfh`→`12926` (0.987), `-71_abcfh`→`2627_b` (0.278), `-71_abcfh`→`9555_b` (0.855).

- **Covers** (coverage, own and as part of the hierarchy): 9 candidate nodes.

Hypothesis. Service-call x low-day-usage gate. 807 = relu(-1.29 - 1.40 dayMin + 0.04 csc + 0.675 g(csc) + 0.622 relu(0.42 - 1.00 eveCharge + 0.36 csc + 0.34 relu(0.99 csc - 0.65) + 0.15 state:41)), with g(csc) = relu(1.01 + 0.28 csc + 0.33 relu(0.86 csc - 0.10)) always active (convex in csc, kink near 2 calls); 1264 = relu(0.08 + 0.36*807 - 0.33 intlCharge). The logit gets +1.60*807 + 1.81*1264. Hypothesis: 807 is a hinge that switches on at LOW day minutes, and its kink moves up with each customer-service call (median kink on train: 147 day minutes at 0 calls, 159 at 1, 174 at 2, 198 at 3, 220 at 4, 244 at 5, 276 at 6); lower evening charge also raises the kink. Below the kink the gate's slope is -3.15 logit per SD of day minutes, which more than cancels the direct +2.04, so the logit is V-shaped in day minutes (net -1.12 per SD below the kink). 1264 adds a second hinge on low international charge that is active on 1281 rows where 807 is off. Evidence (train, n=4000): CSC>=4 and day minutes < 160: 113 customers, empirical churn 0.885, model mean 0.655; CSC<4 and day minutes < 120: 476 customers, empirical 0.046, model 0.084. Mean churn by calls 0-6: 0.115/0.110/0.110/0.106/0.437/0.571/0.679, model 0.098/0.092/0.099/0.147/0.333/0.645/0.923; share of rows with 807 on rises 0.32 -> 0.96. Sensitivity: 807 mostly day minutes (0.64), service calls (0.26), evening charge (0.22). Ablation: mean-ablation AUC 0.764 (full 0.901); least-squares linear replacement 0.848 (R2 0.70/0.75); restored alone onto an all-linearised model 0.898 (all-linear 0.837). This is the model's main nonlinearity.

Closed form:

$$
\begin{aligned}
y_0 = \operatorname{relu}\big(&-1.29 + 0.622\operatorname{relu}\big(0.416 \\
&+ 0.154x_{state:41_b} + 0.342\operatorname{relu}\big(-0.649 \\
&+ 0.987x_{numberCustomerServiceCalls_abcfh}\big) \\
&+ 0.36x_{numberCustomerServiceCalls_abcfh} - 1.0x_{totalEveCharge_b}\big) \\
&+ 0.0405x_{numberCustomerServiceCalls_abcfh} + 0.675\operatorname{relu}\big(1.01 \\
&+ 0.325\operatorname{relu}\big(-0.0966 + 0.855x_{numberCustomerServiceCalls_abcfh}\big) \\
&+ 0.278x_{numberCustomerServiceCalls_abcfh}\big) - 1.4x_{totalDayMinutes_b}\big) \\
y_1 = \operatorname{relu}\big(&0.0759 + 0.363\operatorname{relu}\big(-1.29 \\
&+ 0.622\operatorname{relu}\big(0.416 + 0.154x_{state:41_b} \\
&+ 0.342\operatorname{relu}\big(-0.649 + 0.987x_{numberCustomerServiceCalls_abcfh}\big) \\
&+ 0.36x_{numberCustomerServiceCalls_abcfh} - 1.0x_{totalEveCharge_b}\big) \\
&+ 0.0405x_{numberCustomerServiceCalls_abcfh} + 0.675\operatorname{relu}\big(1.01 \\
&+ 0.325\operatorname{relu}\big(-0.0966 + 0.855x_{numberCustomerServiceCalls_abcfh}\big) \\
&+ 0.278x_{numberCustomerServiceCalls_abcfh}\big) - 1.4x_{totalDayMinutes_b}\big) \\
&- 0.326x_{totalIntlCharge_c}\big)
\end{aligned}
$$

where y_0 is node `807`; y_1 is node `1264`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.0119, and the exact function by at most 8.88×10^{-16}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. Service-call x low-day-usage gate. 807 = relu(-1.29 - 1.40 dayMin + 0.04 csc + 0.675 g(csc) + 0.622 relu(0.42 - 1.00 eveCharge + 0.36 csc + 0.34 relu(0.99 csc - 0.65) + 0.15 state:41)), with g(csc) = relu(1.01 + 0.28 csc + 0.33 relu(0.86 csc - 0.10)) always active (convex in csc, kink near 2 calls); 1264 = relu(0.08 + 0.36*807 - 0.33 intlCharge). The logit gets +1.60*807 + 1.81*1264. Hypothesis: 8

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1.6.10 A10: csc_hinge

Kind: Leaf annotation (depth 2); parent A8 (section 1.6.8).

- **Entry nodes:** `-71_d` (numberCustomerServiceCalls_d), `-45_c` (state:49_c).
- **Exit nodes:** `7208`.
- **Internal nodes:** `17032`.
- **Edges (weight):** `17032`→`7208` (0.544), `-71_d`→`17032` (1), `-45_c`→`7208` (-0.359).
- **Covers** (coverage, own and as part of the hierarchy): 3 candidate nodes.

Hypothesis. 7208 = relu(-0.67 + 0.54 relu(csc + 0.008) - 0.36 state:49), logit weight +1.48. Hypothesis: a hinge on customer-service calls that switches on from 4 calls upward (kink at 3.17 calls) and adds about +0.61 logit per extra call (contribution 0.60 at 4 calls, 1.22 at 5, 1.83 at 6, 2.45 at 7); state:49 switches it off (12 state:49 customers with 4+ calls, 5 of them churned). Active on 306 of 318 customers with 4+ calls. Evidence: line plot of 7208 on train rises from 0 to 2.5 (correlation with calls 0.66); mean ablation AUC 0.8996, linear replacement 0.8986; restored alone onto the all-linear model 0.851 (all-linear 0.837), but redundant once the gate is present (gate alone 0.898).

Closed form:

$$y_0 = \mathrm{relu}\big(-0.673 + 0.544\,\mathrm{relu}\big(0.00784 + 1.0x_{numberCustomerServiceCalls_d}\big) - 0.359x_{state:49_c}\big)$$

where y_0 is node `7208`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.00181, and the exact function by at most 4.44×10^{-16}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. 7208 = relu(-0.67 + 0.54 relu(csc + 0.008) - 0.36 state:49), logit weight +1.48. Hypothesis: a hinge on customer-service calls that switches on from 4 calls upward (kink at 3.17 calls) and adds about +0.61 logit per extra call (contribution 0.60 at 4 calls, 1.22 at 5, 1.83 at 6, 2.45 at 7); state:49 switches it off (12 state:49 customers with 4+ calls, 5 of them churned). Active on 306 of 318 cust

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

1.7 Audit against the definitions

Automatic findings on the stored explanation (codes as in the JSON record's `issues`); they flag what an examiner should check, not verdicts.

- WARNING `unverified`: No human verification is recorded for this explanation.

end of thesis draft